A Survey on Image Restoration in the Foundation Model Era

Jingbo Lin, Renlong Wu, Huangyao Zhang, Yichi Tang, and Wangmeng Zuo

***Abstract*—Foundation models provide transferable representations, generative priors, and multimodal understanding, offering new ways to address diverse image degradations and restoration requirements. This survey reviews three complementary directions: leveraging existing foundation models for image restoration, developing restoration-oriented foundation models, and coordinating restoration models and tools through agents. We examine how pretrained representations, generative priors, and additional supervision support image restoration. For restoration-oriented foundation models, we compare pretraining, unified modeling, and adaptation strategies across tasks and domains. For agentic image restoration, we examine how agents coordinate restoration models and tools within a unified framework for diverse tasks through task planning, tool selection, feedback-driven refinement, and experience retrieval. We summarize reported benchmark results and model budgets to examine the strengths and limitations of existing methods, taking into account differences in training coverage, adaptation, and evaluation settings. Finally, we discuss challenges and future directions in generalization and adaptation to real-world degradations, faithful and user-controllable restoration, reliable assessment for agents, and efficient and reproducible deployment. A curated list of works reviewed in this survey is available at https://github.com/mrluin/Awesome-IR-in-FM-era.**

***Index Terms*—Image Restoration, Foundation Models, Generative Priors, Agentic Restoration.**

I. Introduction

Image restoration aims to recover high-quality images from observations affected by noise, blur, low resolution, or other degradations. In real-world scenarios, these degradations may coexist or vary spatially, while restoration objectives also depend on the intended use of the result. For example, old photo restoration emphasizes visual quality, aiming to produce clear and natural-looking results [1]. Restoration for autonomous driving should preserve scene information needed for downstream perception [2]. In interactive photo enhancement, user instructions specify the desired changes and guide the restoration process [3], [4].

Foundation models are pretrained on large-scale data and can be adapted to a range of downstream tasks [5]. For image restoration, pretrained visual and vision-language representations provide semantic and structural information for image reconstruction [6]–[8]. Generative priors guide the inference of image details that are missing or ambiguous in degraded observations [9]–[11]. Language and multimodal capabilities support the interpretation of image content and user instructions, enabling natural-language specifications of desired changes to guide restoration [3], [4], [12].

Related surveys focus on universal image restoration [13], language-driven restoration with semantic-aware quality assessment [14], and restoration agents [15]. This survey examines the use and development of foundation models in image restoration through three successive parts covering *image restoration with foundation models*, *restoration-oriented foundation models*, and *agentic image restoration*. This organization progresses from leveraging pretrained capabilities for reconstruction, supervision, and quality assessment to restoration-oriented pretraining and adaptation, and then to the coordination of restoration models and tools through agents. Fig. 1 summarizes representative developments. Fig. 2 presents the taxonomy, while Fig. 3 illustrates representative mechanisms.

In image restoration with foundation models, visual and vision-language representations complement features learned by restoration networks with semantic and structural information [7], [8]. Pretrained generative models contribute image priors that can be combined with degraded observations through inference-time constraints [16] or learned conditioning [10], [11]. Natural-language instructions specify restoration requirements and desired operations [3]. For training supervision, pretrained models generate degraded inputs paired with clean targets [17] or provide diffusion-based losses for separate restoration networks [18]. Vision-language and multimodal models assess image quality through image-text correspondence [19] or quality-related instruction tuning [20].

The development of foundation models for image restoration involves restoration-oriented pretraining and downstream adaptation. Related efforts in universal restoration and multi-task pretraining investigate shared representations and priors for downstream restoration tasks [21], [22]. Masked reconstruction learns clean content from partially observed degraded inputs [23], while degradation classification supervises restoration-encoder pretraining [24]. Large-scale generative training aims to broaden restoration coverage through diverse image data [25]. Depending on their design and training, pretrained models can be applied directly to supported restoration tasks or adapted through task-specific fine-tuning. Task-specific adapters provide a parameter-efficient option for such adaptation [26]. Task requirements can also be specified through model inputs. Visual in-context learning uses input-output examples to specify the desired transformation [27], [28]. Multimodal conditioning provides further ways to express low-level processing requirements [29].

Agentic restoration coordinates restoration models and tools according to the input image and restoration target. Language and multimodal models help interpret image conditions and user requirements to inform tool selection [30], [31]. Planning, search, and learned policies organize restoration sequences, with actions determined in advance or selected based on updated observations [32]–[34]. Existing systems also employ retrieved experience to guide action selection [35]. The selected tools, which may include instruction-guided generative

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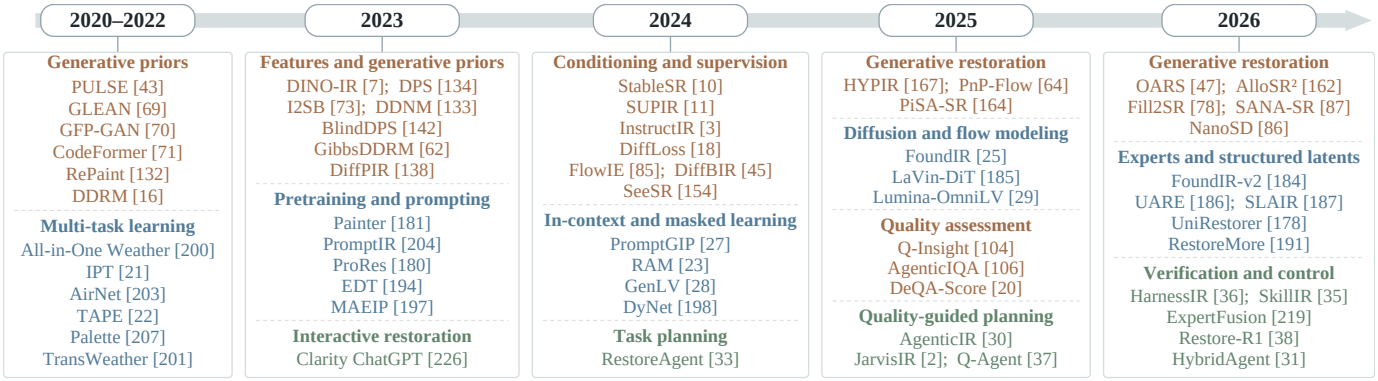


Fig. 1. Representative developments in image restoration across 2020–2026. Orange, blue, and green indicate image restoration methods with foundation models, restoration-oriented foundation model, and agentic image restoration methods, respectively.

foundation models, carry out these operations [36]. At inference, quality assessment can guide candidate acceptance, further operations, or stopping [35], [37]. During training, quality-based rewards can optimize tool-use policies [34], [38].

Sections II, III, and IV discuss these three parts, respectively. Section V summarizes reported benchmark results and model sizes under their respective evaluation settings. Section VI discusses generalization and adaptation to real-world degradations, faithful and user-controllable restoration, reliable assessment for agentic restoration, and efficient and reproducible restoration agents. Section VII concludes the survey.

II. Image Restoration with Foundation Models

Foundation models and related pretrained priors provide visual features, segmentation priors, text conditions, image priors, training resources, and quality judgments. Table I summarizes representative methods.

A. Image Restoration with Vision and Language Models

Pretrained vision and language models provide visual and semantic information to guide image restoration. Visual features and segmentation masks describe content and spatial regions. Vision-language models (VLMs) relate images to text, while multimodal large language models (MLLMs) describe content and degradation. Text encoders embed requested operations, while generative language models can interpret or formulate instructions.

1) Visual Features and Segmentation Priors: Pretrained visual features and segmentation priors provide content and spatial guidance for restoration. DINO-IR [7] fuses multiscale DINOv2 and restoration features with contrastive supervision. RAESR [108] learns a generative decoder from frozen DINOv3-L features without diffusion sampling. GramSR [109] instead conditions SD2.1 on DINOv3 features through pixel, semantic, and texture low-rank adaptation (LoRA) [110] branches. CLD transfers sRGB-pretrained semantic representations into a RAW denoiser through cross-domain feature fusion [111]. DiffLI^{2}D transfers bottleneck features from a frozen ImageNet-pretrained diffusion model to a supervised restorer without reverse sampling [112]. For segmentation guidance, SAM prior tuning [39] uses trainable mask-conditioned units with SAM and the restorer fixed. SAM-Deblur [113] pools regional features, while PPT-former [114] encodes SAM parsing maps into multiscale conditions and fuses them through attention and gated feed-forward processing. Precomputed maps remain inference inputs. Distillation provides another way to transfer pretrained guidance during training. SAM distillation [40] refines initial reconstructions with a mask-guided branch. This branch supervises the original branch through output and semantic-relation losses. PTG-RM [115] guides training refinement with pretrained features. Both remove auxiliary teachers after training.

2) Degradation and Instruction Conditioning: Information about the input degradation can help determine how the image should be restored. In VL-UR, weather descriptors may be selected automatically or supplied by the user [116]. DA-CLIP [8] learns a contrastive controller around fixed CLIP encoders for content and degradation conditions, while LoRA-IR [117] routes low-rank restoration experts. ClearAIR [118] combines DeQA quality and SAM regional cues. MoFE-MLLM [119] injects Qwen2.5-VL features into a restoration encoder and decoder and routes frequency experts. For feature aggregation, QuReC [120] matches spatial queries to fixed CLIP text prototypes. MGN-AIR [121] uses text and learned visual prompts to vary pixel-level processing, while SLER-IR [122] routes layer-wise experts with distinct content and degradation representations.

Natural-language instructions can specify the restoration operations requested by the user. InstructIR [3] uses generated training instructions and projects frozen BGE embeddings to modulate channels in a NAFNet-based restorer. TextPromptIR [123] instead fine-tunes BERT. Image-based conditions can complement text cues. Generated references supply content, while degraded-clean examples indicate the degradation to remove. LMDIR [41] generates SDXL [124] content references from content and degradation descriptions. TextDegRemoval [42] maps CLIP image features into continuous textual embeddings and learns a textual restoration module to remove degradation information. The corrected embeddings condition pretrained Stable Diffusion to generate an auxiliary guidance image, whose features are integrated into a restoration backbone through coarse-to-fine matching

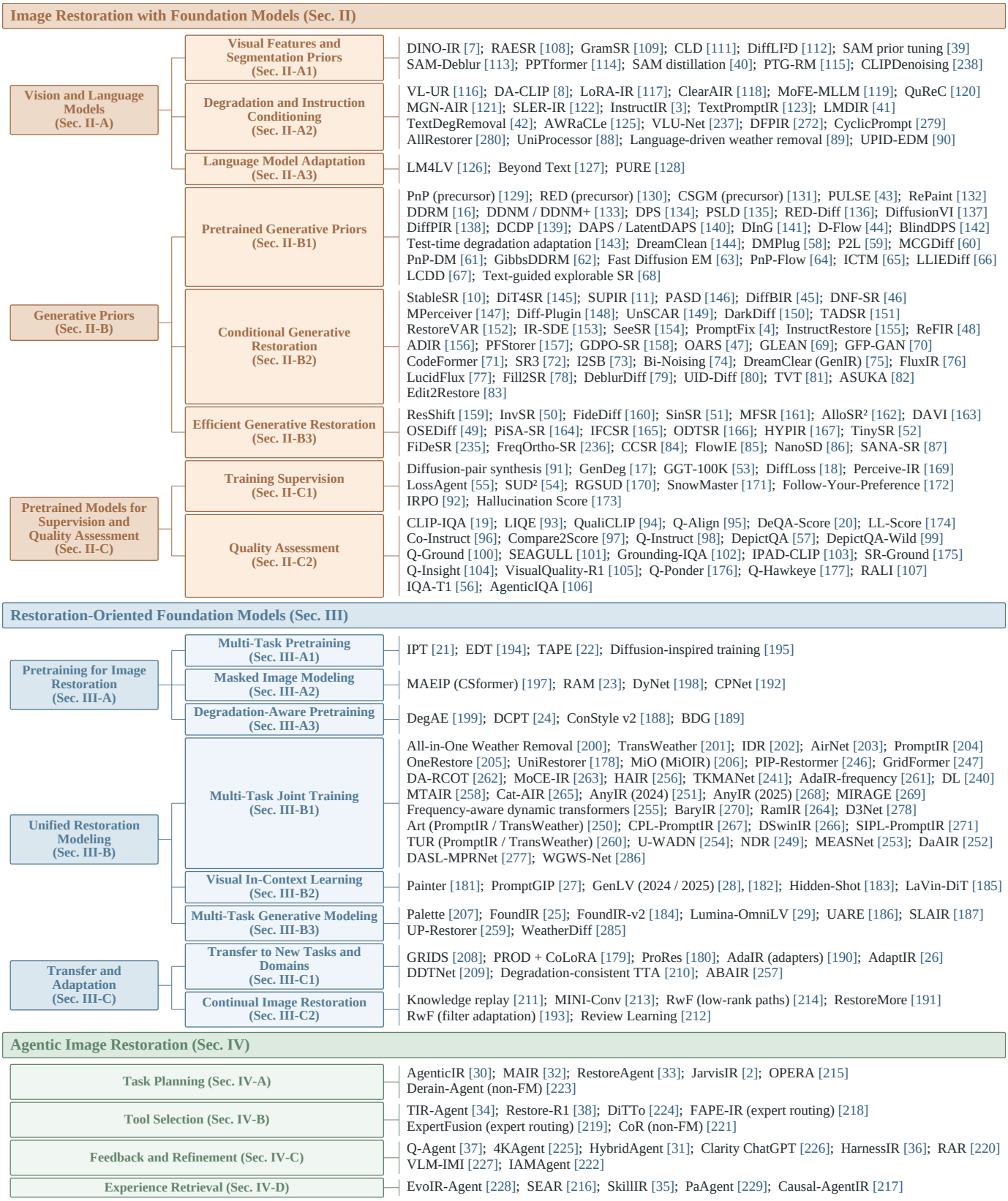


Fig. 2. Taxonomy of image restoration methods in the foundation model era.

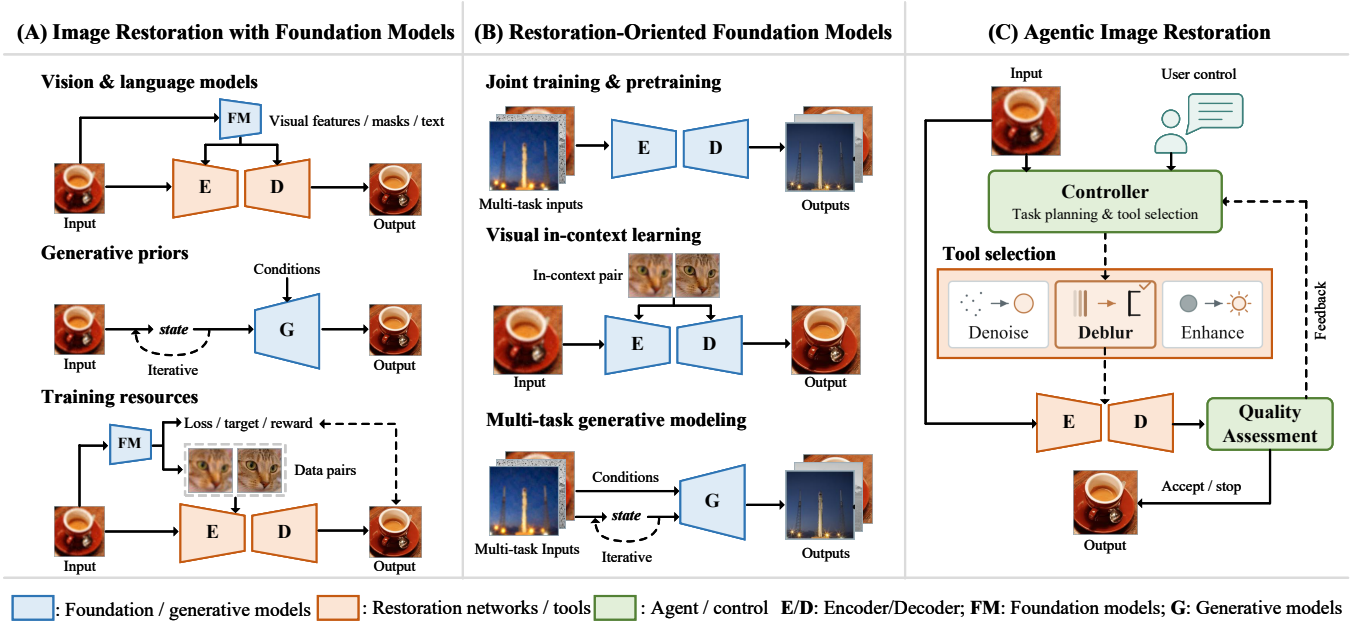


Fig. 3. Representative mechanisms for (A) restoration with foundation models, (B) restoration-oriented foundation models, and (C) agentic restoration.

and multiscale aggregation. AWRaCLe [125] instead extracts CLIP features from a degraded-clean context pair and injects a degradation representation at decoder levels for queries sharing that degradation.

UniProcessor combines a degradation subject prompt with a manipulation prompt to specify the degradation and the requested restoration operation. Multimodal context controls are injected into the restoration network through cross-attention [88]. Language-driven adverse-weather removal uses a VLM-derived degradation prior to select convolutional experts according to the corruption type and severity [89]. For unpaired deraining, UPID-EDM learns domain-representation prompts for rainy and clean images with frozen CLIP encoders. The learned prompts define a rain-relevance discarding energy function, while CLIP image features define a rain-irrelevance preserving energy function. These two terms guide reverse sampling in a diffusion model pretrained on clean images [90].

3) Language Model Adaptation: Beyond conditioning, pretrained language models can predict image tokens through learned image-token mappings. LM4LV [126] first adapts an MAE decoder. It then freezes visual modules and LLaMA while learning linear adapters and task tokens for autoregressive continuous-token prediction. BeyondText [127] keeps the language backbone fixed and learns an image tokenizer and decoder using its discrete vocabulary tokens. Input-output examples specify visual transformations. In contrast, PURE instruction-tunes Lumina-mGPT to generate degradation labels, semantic descriptions, and restored image tokens [128]. These methods extend pretrained language models from providing restoration guidance to directly predicting image representations, either by learning visual interfaces around a frozen language backbone or by instruction-tuning a multimodal model for restoration.

B. Image Restoration with Generative Priors

Generative models provide learned image priors that help resolve ambiguity in degraded observations, complementing the visual and language guidance discussed above. We review how fixed generative priors guide restoration, how conditioning and model adaptation support generative restoration, and how efficient designs reduce inference cost. We focus on the utilization of pretrained generative models and also discuss task-specific generative models to explain their restoration designs.

1) Pretrained Generative Priors: Early work on image inverse problems explored how denoisers and pretrained generators could serve as image priors. Plug-and-play reconstruction incorporates a denoiser into an iterative optimization algorithm [129], whereas regularization by denoising (RED) introduces a denoiser-dependent regularization functional [130]. Compressed sensing with generative models instead optimizes a latent code within a pretrained generator to fit linear measurements [131].

With pretrained GANs and diffusion models, image priors can be incorporated into restoration through latent optimization or observation-conditioned sampling. The model parameters remain fixed, while latent inputs, image estimates, or sampling states are adjusted to match the degraded observation. PULSE optimizes a GAN latent code for downsampling consistency [43]. RePaint conditions on known pixels and resamples the trajectory [132]. DDRM [16] uses spectral conditioning, while DDNM [133] separates measured range-space components from generated null-space content. DDNM+ handles noisy observations. For differentiable degradation operators, DPS [134] approximates likelihood gradients through denoised estimates under its noise assumptions. For linear inverse problems, PSLD [135] propagates gradients through a latent decoder. Its gluing guidance matches clean latent

Table I. Representative methods that use foundation models for image restoration.

Method	FM / Backbone	Role in restoration	Guidance information
DINO-IR [7]	DINOv2	Semantic guidance	Multiscale DINOv2 features
SAM priors [39]	SAM + restorer	Segmentation guidance	SAM masks
SAM distillation [40]	SAM + teacher restorer	Semantic-prior distillation	SAM masks (training only)
DA-CLIP [8]	CLIP ViT-B/32	Image conditioning	Content and degradation embeddings
InstructIR [3]	BGE + NAFNet	Instruction-guided restoration	Natural-language instructions
LMDIR [41]	GPT-4o + CLIP + SDXL	Reference-guided restoration	Content, degradation text, and references
TextDegRemoval [42]	CLIP + Stable Diffusion	Guidance-image generation	Restored text embeddings
DDRM [16]	ADM or DDPM	Posterior sampling	Measurements, linear operator, and noise
PULSE [43]	StyleGAN	Latent optimization	LR observation and downscaling loss
D-Flow [44]	Cond-OT or latent T2I flow	Controlled generation	Measurements and class or text conditions
DiffBIR [45]	Stable Diffusion 2.1	Conditional restoration	Pre-restored image and IRControlNet
DNF-SR [46]	FLUX-Kontext	Generative super-resolution	LR latents, noisy LR latents, and captions
OARS [47]	Qwen-Image-Edit	Generative super-resolution	LR image and enhanced prompt
ReFIR [48]	SD2 or SDXL	Retrieval-guided restoration	LQ image and retrieved HQ references
OSEDiff [49]	Stable Diffusion 2.1	One-step super-resolution	LQ latent and degradation prompts
InvSR [50]	SD-Turbo	Generative super-resolution	LR-derived initialization and fixed prompt
SinSR [51]	ResShift teacher	Restoration distillation	LR image and perturbed latent (training)
TinySR [52]	SD3 + VAE	Generative super-resolution	LR latent
GGT-100K [53]	Nano Banana 2 + Gemini	Training-target generation	LQ image and adaptive prompt
DiffLoss [18]	ImageNet ADM	Diffusion-based supervision	Noised restored and reference images
SUD^2 [54]	CelebAMask-HQ DDPM	Pseudo-target generation	Noised reconstructions (training)
LossAgent [55]	Llama-3-8B-Instruct	Loss-weight optimization	Feedback and weight history (training)
CLIP-IQA [19]	CLIP	No-reference assessment	Image and antonym prompts
DeQA-Score [20]	mPLUG-Owl2	No-reference assessment	Image and quality question
IQA-T1 [56]	Qwen3-VL-4B	No-reference assessment	Image and tool-generated evidence
DepictQA [57]	CLIP + Vicuna-7B	Description and comparison	Images, question, and optional reference
DMPlug [58]	Diffusion prior	Initial-noise optimization	Measurements and forward operator
P2L [59]	Latent diffusion prior	Prompt and latent optimization	Measurements and continuous text conditions
MCGDiff [60]	Diffusion prior	Sequential Monte Carlo	Linear measurements and particle weights
PnP-DM [61]	Diffusion prior	Split Gibbs sampling	Likelihood and denoising conditionals
GibbsDDRM [62]	Image diffusion prior	Blind inverse problems	Linear measurements and operator prior
Fast Diffusion EM [63]	Diffusion + kernel denoiser	Monte Carlo EM	Measurements and learned kernel prior
PnP-Flow [64]	Flow-matching prior	Plug-and-play restoration	Forward model and time schedule
ICTM [65]	Flow prior	Trajectory matching	Linear measurements and local objectives
LLIEDiff [66]	Stable Diffusion	Low-light posterior sampling	Illumination and measurement constraints
LCDD [67]	Image diffusion prior	Controllable denoising	Noise level and endpoint blending weight
Text-guided explorable SR [68]	Generative priors	Explorable super-resolution	LR observation, operator, and text
GLEAN [69]	GAN latent bank	Conditional reconstruction	LR features and latent-bank features

estimates to encoded observed and generated components.

Within this fixed-model setting, optimization can be performed at different stages of the generative process. DMPlug treats reverse diffusion as a differentiable mapping and optimizes its initial noise to fit the observation [58]. P2L instead tunes continuous text conditions during reverse sampling, alternating prompt updates with latent and pixel-level estimation while retaining the pretrained diffusion weights [59].

Image estimates can be updated through regularization or separate measurement and prior steps. RED-Diff combines measurement matching with fixed diffusion-denoiser regularization [136]. For denoising with unknown noise statistics, DiffusionVI estimates noise-precision posteriors and intermediate maximum a posteriori images using a fixed unconditional diffusion prior [137]. DiffPIR [138] combines diffusion sampling with half-quadratic splitting of data fitting and prior regularization. DCDP [139] and DAPS [140] decouple measurement updates from purification. DInG [141] instead decouples its likelihood proxy from the current transition, obtaining Gaussian posterior updates without denoiser backpropagation.

A further distinction concerns how posterior samples are constructed. MCGDiff uses sequential Monte Carlo with weighted particles for intermediate linear inverse problems [60]. PnP-DM instead uses split-Gibbs sampling to alternate likelihood sampling and Gaussian-denoising posterior sampling through a pretrained diffusion model [61].

Pretrained flow models provide an alternative to diffusion priors through a learned velocity field. D-Flow integrates this fixed field and optimizes the initial noise for observation consistency [44]. PnP-Flow instead derives time-dependent denoising from a fixed velocity field and interleaves it with data-fidelity updates and interpolation toward the flow path [64]. Unlike differentiation through the integrated D-Flow trajectory, the local operations in PnP-Flow avoid backpropagation through the generative network or ODE. Iterative Corrupted Trajectory Matching formulates inference for linear inverse problems through local objectives along a flow trajectory [65].

When the degradation operator is unknown, blind restoration can infer it jointly with the image. BlindDPS uses separately trained diffusion priors for degradation operators such as blur kernels [142]. GibbsDDRM instead uses a partially collapsed Gibbs formulation with an operator prior to alternate inference of the image and an unknown linear operator [62]. Fast Diffusion EM uses sample-based expectation maximization for deconvolution, with a trained blur-kernel denoiser regularizing its maximization step [63]. Its fast variant interleaves kernel updates with approximate image inference along a reverse trajectory.

Table I. Representative methods that use foundation models for image restoration. Continued.

Method	FM–Backbone	Role in restoration	Guidance information
GFP-GAN [70]	Pretrained face GAN	Blind face restoration	Restoration features and facial prior
CodeFormer [71]	Facial codebook	Code prediction	LQ tokens and fidelity control
SR3 [72]	Task-trained diffusion	Iterative refinement	LR image
I^2SB [73]	Task-trained diffusion bridge	Schrödinger bridge	Paired training boundaries and degraded input
Bi-Noising [74]	Conditional diffusion + prior	Re-noising and refinement	Task condition and unconditional prior
DreamClear [75]	PixArt-α + ControlFormer	Conditional restoration	LQ image, pre-restored proxy, and caption
FluxIR [76]	FLUX + MM-DiT adapter	Conditioning adaptation	Degraded latent and embedding offsets
LucidFlux [77]	FLUX.1 + SigLIP	Caption-free conditioning	LQ image, proxy, and fixed instruction
Fill2SR [78]	FLUX-Fill + LoRA	Inpainting-based SR	LQ latent and full mask (Redux optional)
DeblurDiff [79]	Stable Diffusion + control	Adaptive convolution	Spatially varying latent kernels
UID-Diff [80]	Stable Diffusion + Q-Formers	Deblurring adaptation	Structure and blur representations
TVT [81]	Adapted VAE + diffusion U-Net	Fine-structure preservation	Compatible $4\times$-compression latents
ASUKA [82]	MAE + inpainting generator	Image inpainting	Known pixels and MAE reconstruction features
Edit2Restore [83]	FLUX.1 Kontext + LoRA	Few-shot adaptation	Task prompts and degraded images
CCSR [84]	Stable Diffusion + ControlNet	Content refinement	LQ image and trained decoder refinement
FlowIE [85]	Adapted diffusion U-Net	Conditional rectified flow	Preliminary restoration and spatial controls
NanoSD [86]	Compressed SD1.5 + VAE	Generative backbone	Distillation and hardware profiling (training)
SANA-SR [87]	Autoencoder + LinearDiT	One-step restoration	Image latents and prompt-aware pruning
UniProcessor [88]	Q-Former + text encoder	Instruction-guided processing	Subject and manipulation prompts
Language-driven weather removal [89]	VLM + convolutional experts	Expert selection	Weather type and severity
UPID-EDM [90]	CLIP + clean-domain diffusion	Energy-informed deraining	Domain prompts and content energies
Diffusion-pair synthesis [91]	Real-LQ-trained DDPM	Training-pair synthesis	Clean image and simulated degradation
IRPO [92]	AdaIR + Qwen-VL	GRPO post-training	Expert and task rewards with paired targets
LIQE [93]	Adapted CLIP	Quality scoring	Scene, distortion, and quality templates
QualiCLIP [94]	Adapted CLIP	Opinion-unaware alignment	Image and quality-antonym prompts
Q-Align [95]	Multimodal language model	Quality-level prediction	Image and text-defined rating levels
Co-Instruct [96]	Instruction-tuned MLLM	Quality comparison	Candidate images and comparison query
Compare2Score [97]	Adapted MLLM	Comparison-based scoring	Image and quality-scale anchors
Q-Instruct [98]	Instruction-tuned MLLM	Visual question answering	Image and appearance query
DepictQA-Wild [99]	Adapted multimodal model	Descriptive assessment	Image, query, and optional reference
Q-Ground [100]	MLLM + mask decoder	Quality grounding	Image and quality text
SEAGULL [101]	Vision-language ROI model	Region quality assessment	Image and region mask
Grounding-IQA [102]	Multimodal language model	Localized quality assessment	Image and local quality question
IPAD-CLIP [103]	Adapted CLIP	Local artifact detection	Image and artifact-aware features
Q-Insight [104]	RL-adapted MLLM	IQA and degradation perception	Image and quality query
VisualQuality-R1 [105]	RL-adapted MLLM	Quality ranking	Image and assessment prompt
AgenticIQA [106]	Adapted MLLM + IQA tools	Assessment orchestration	Image, query, and optional reference
RALI [107]	Image encoder	Quality prediction	Reasoning-aligned image features

Unknown degradations can also be handled by adapting the degradation mapping to the test image. Test-time degradation adaptation [143] learns an adapter and discriminator from the test input during denoising while keeping the diffusion prior fixed. The learned degradation mapping provides observation guidance. DreamClean takes a different route, avoiding an explicit degradation operator through diffusion inversion and refinement [144].

In low-light enhancement and denoising, inference can be tailored to illumination and noise characteristics. LLIEDiff models low-light degradation through an exposure-adaptive illumination map and constrains latent sampling with measurement and latent regularization, without paired task training [66]. For Gaussian denoising, LCDD matches the input noise level to a diffusion state and blends the resulting one-step and multistep estimates [67].

Observation constraints can also be combined with semantic control over the restored image. Text-guided explorable super-resolution combines a known downsampling constraint with semantic guidance in pretrained generative models [68]. Language steers inference toward alternative high-resolution explanations of the same low-resolution observation, without training a restoration-specific conditioning interface.

2) Conditional Generative Restoration: Learned conditional interfaces provide another way to connect degraded observations to generative priors. Conditional generative restoration uses degraded-image structure, text, or reference images to guide reconstruction, with methods differing in the conditioning modules or generator components they train or adapt.

Early learned restoration interfaces draw on GAN priors or discrete codebooks. GLEAN learns to retrieve and combine information from a pretrained generative latent bank for super-resolution [69]. GFP-GAN connects restoration features to a pretrained face generator through channel-split spatial feature transforms [70]. CodeFormer instead learns a discrete facial codebook and decoder, then fixes them while training low-quality encoding and transformer code prediction [71]. Its controllable feature transfer adjusts the balance between input fidelity and visual quality.

With pretrained diffusion models, degraded-image features can be injected through learned encoders and attention modules. StableSR [10] trains time-aware encoding and feature modulation to condition frozen Stable Diffusion on degraded-image structure. Its optional feature wrapping transfers VAE-encoder features into decoder features. DiT4SR [145] adds a low-resolution stream to pretrained multimodal transformer

blocks for joint attention with noisy-image tokens. Since degraded inputs contain both corruption and scene structure, several methods seek more degradation-robust conditions. SUPIR [11] adapts a degradation-robust encoder and restoration adapter. PASD [146] trains a degradation-removal module to produce multiscale control features for pixel-aware cross-attention. DiffBIR [45] trains IRControlNet on preliminary restorations to condition frozen Stable Diffusion. During sampling, SUPIR interpolates predictions toward the input latent, while DiffBIR optionally applies spatially weighted guidance toward the preliminary estimate.

Restored proxy images can supplement degraded-image conditions with cleaner structural or semantic features. DreamClear uses a PixArt-α prior with ControlFormer, combining degraded-image and SwinIR-pre-restored streams through token-wise adaptive experts [75]. Captions provide semantic conditions. LucidFlux trains dual image-conditioning branches and a connector that maps proxy-image SigLIP features into the conditioning space of a frozen Flux.1 model [77]. Its conditions vary across timesteps and layers. A fixed restoration instruction is used instead of image-specific captions.

Direct conditioning on degraded latents can use learned image-text adapters or native masked-image slots. FluxIR processes degraded latents and learned text-embedding offsets in an MM-DiT adapter and broadcasts the resulting conditions across its pretrained FLUX backbone [76]. Its image and text interfaces are trained jointly for restoration. Fill2SR uses the native masked-image slot of FLUX-Fill: a full-image mask and encoded low-quality image provide the restoration condition, with LoRA updates learning the conditional flow [78]. Optional Redux tokens add semantic guidance.

The conditioning interface can also reflect the structure of a particular restoration task. For motion deblurring, DeblurDiff repeatedly predicts spatially varying latent kernels and uses adaptive filtering to update structural guidance for Stable Diffusion [79]. UID-Diff separates structure and blur through two Q-Formers, combining synthetic paired deblurring with unpaired target-domain blur transfer and reconstruction [80].

Inpainting shifts the conditioning problem from correcting corrupted content to completing missing regions. ASUKA aligns an MAE reconstruction prior with a frozen generative model in place of text conditioning [82]. It uses observed-region conditions to support continuity across masked and unmasked areas and learns an inpainting-specialized decoder for local color harmonization.

To support multiple degradations and tasks, restoration models can select or share learned conditioning components. MPerceiver [147] weights learned prompt pools by CLIP-based degradation probabilities. Its learned branches supply global text conditions and multiscale visual features. Diff-Plugin [148] trains task plugins independently. In contrast, UnSCAR [149] learns shared control and expert components to condition a frozen diffusion backbone across 16 jointly trained tasks.

Beyond conditioning modules, restoration adaptation can modify image encoding and decoding. DarkDiff [150] adapts a residual VAE and region-conditioned latent denoiser to map noisy linear RGB to sRGB. TADSR [151] uses time-aware encoding to vary restoration with the inference timestep. Transfer VAE Training (TVT) reduces the VAE spatial compression used by a pretrained diffusion U-Net, first training a compatible decoder and then its encoder with the decoder fixed [81]. Added high-resolution boundary layers are fully fine-tuned, while low-rank adaptation connects the original U-Net to the new representation. The reduced compression targets fine structures that coarse encoding may discard.

The generative formulation can also be tailored to restoration, from autoregressive token prediction to conditional diffusion processes. RestoreVAR [152] predicts clean discrete indices scale by scale, then uses a non-generative refiner to obtain continuous decoder features. SR3 learns iterative refinement conditioned on a low-resolution input [72]. IR-SDE learns the reverse of a process whose mean is the degraded image [153], while I^2SB learns an image-to-image Schrödinger bridge between paired endpoint distributions [73]. Bi-Noising combines a task-trained conditional diffusion model with an unconditional prior through re-noising and refinement [74].

Alongside image conditions, text can describe image content or specify requested changes. SeeSR [154] aligns low- and high-quality tags and features, then supplies object tags and continuous conditions. PromptFix [4] combines frozen-VLM descriptions with user instructions and updates decoding LoRAs during sampling. InstructRestore [155] trains a mask decoder and ControlNet around frozen Stable Diffusion for instructions inside and outside target regions.

Pretrained editing models offer an initialization for learning restoration operations. DNF-SR [46] uses negative-aware post-training of FLUX-Kontext with unnoised and noise-perturbed inputs. Edit2Restore trains LoRA updates for FLUX.1 Kontext using small paired task datasets and simple operation prompts [83]. It uses 32-128 paired examples per task and also studies a shared adapter trained across five restoration tasks.

Besides text, reference images can supply restoration guidance or adaptation examples. ReFIR [48] retrieves relevant images and injects textures through cross-image attention and spatial control without retraining. ADIR [156] retrieves adaptation examples for LoRA updates before measurement-guided reconstruction. PFStorer [157] learns identity modules from references once per person for later images.

Restoration preferences can also be incorporated through reward-based training. GDPO-SR [158] uses regional rewards, while OARS [47] applies reinforcement learning to Qwen-Image-Edit with COMPASS training rewards and retains iterative generation.

3) Efficient Generative Restoration: Generative restoration can be accelerated by shortening sampling trajectories or reducing the computational cost of model components. Trajectory design, initialization, and learned one- or few-step mappings address sampling cost, while compression targets the generative backbone and autoencoder.

Trajectory design and initialization reduce the iterative work needed to connect degraded inputs to restored images. ResShift [159] learns residual-shifting trajectories, whereas InvSR [50] uses a trained noise predictor to initialize frozen diffusion at an intermediate state, with adjustable sampling

steps. FideDiff [160] learns cross-time consistency by matching different blur levels to a shared sharp target, with a separate Kernel ControlNet supplying blur-kernel conditioning and adaptive timestep prediction. CCSR [84] separates content refinement from detail enhancement, using non-uniform and truncated diffusion sampling for the former and a separately fine-tuned VAE decoder for the latter. FlowIE [85] instead adapts a pretrained diffusion U-Net into a conditional rectified-flow model, with coarse restoration providing guidance for few-step generation.

One- or few-step mappings can also be learned through teacher distillation or trajectory-consistency objectives. SinSR [51] distills the ResShift trajectory into one step, while MFSR [161] learns interval-average velocity with an end-time embedding from a frozen DiT4SR teacher. AlloSR^2 [162] adapts FLUX with LoRA, trajectory consistency, and allomorphic matching, using SNR initialization for one-step inference.

Reconstruction losses and score-based supervision support learning one- or few-step restoration mappings. DAVI [163] learns a measurement-conditioned approximate posterior sampler from a fixed diffusion prior, replacing the sampling trajectory with one forward pass. OSEDiff [49] trains encoder and denoiser LoRAs with reconstruction and score distillation for one denoising step. PiSA-SR [164] learns pixel then semantic LoRAs, fixing the pixel branch during perceptual and classifier score distillation. Default inference merges the two LoRAs, whereas adjustable guidance combines network outputs. IFCSR [165] instead interpolates between fidelity- and realism-oriented output images, enabling further trade-off adjustment without additional network inference.

Adversarial supervision and staged adaptation provide further training designs for single-pass restoration. ODTSR [166] trains Qwen-Image control-noise LoRA with a training-only Wan2.1 discriminator. HYPIR [167] separates degradation removal from latent restoration: it first adapts its VAE encoder, then fixes encoding and decoding while training LoRA in a diffusion-initialized restorer for a single latent forward pass.

Model and representation compression complement sampling acceleration by reducing the computation of the generative backbone and autoencoder. TinySR [52] prunes SD3, compresses the autoencoder, and removes redundant fixed-prompt conditions. NanoSD [86] co-compresses the SD1.5 U-Net and VAE using hardware-profiled architectural changes and generative distillation. Its compact pretrained components can then support downstream restoration frameworks. SANA-SR [87] combines a more compressed latent representation and a LoRA-adapted LinearDiT with one-step restoration and structured pruning.

C. Pretrained Models for Supervision and Quality Assessment

Pretrained models support restoration by constructing training supervision and assessing image quality. These roles connect data preparation and model optimization with the evaluation of restoration outputs.

1) Training Supervision: Training supervision uses pretrained models to construct examples and targets, define differentiable losses, and provide preference or reward signals. Training-pair synthesis can model the degraded side of a restoration pair. Yang et al. [91] train a DDPM [168] on real low-quality images and use it to refine a handcrafted degradation of a clean source image after noise injection. The generator constructs pairs offline, with only the trained restorer used at inference. GenDeg [17] synthesizes controlled degradations while correcting correspondence with clean targets. Fill2SR extracts degradation descriptors from unpaired real images and transfers the degradations onto clean targets using frozen models, producing offline restoration training pairs [78].

Clean-image synthesis can instead expand the sources or targets available for training. The GenIR pipeline in DreamClear generates and filters clean training images before restoration training [75]. GGT-100K [53] retains real degraded inputs and generates candidate clean targets using automatic checks, VLM refinement, and manual verification.

Beyond constructing examples, pretrained models define training objectives through their representations and predictions. DiffLoss [18] compares noise predictions and bottleneck features of restored and reference images under a fixed diffusion model, retaining only the trained restorer at inference. Perceive-IR [169] first learns CLIP quality prompts, then fixes them for losses while retaining DINOv2 conditions at inference. LossAgent [55] uses an LLM to select differentiable-loss weights between training stages.

Pretrained predictors also provide supervision through pseudo-targets. For unpaired measurements, SUD^2 [54] denoises current reconstructions into fixed pseudo-targets without backpropagating through the denoiser, retaining only the trained reconstruction network at inference. RGSUD [170] uses training-only DACLIP-IQA rewards for deraining pseudo-pairs. SnowMaster [171] preference-tunes an assessor for updating desnowing pseudo-labels.

Quality judgments can enter optimization as preference pairs or rewards. Follow-Your-Preference uses reward-model preference pairs and direct preference optimization to adapt BrushNet branches or selected FLUX Fill attention weights [172]. IRPO applies GRPO to a stochastic control policy in an AdaIR-based restorer using generic, Qwen-VL expert, and task-specific rewards, while jointly updating the backbone with paired reconstruction and consistency supervision [92]. Hallucination Score [173] uses a differentiable reward combining clean-reference DINO or CLIP similarity with MUSIQ quality scores. Its GPT-4o fidelity assessment also uses a clean reference.

2) Quality Assessment: Quality assessors supply training signals and also evaluate restoration outputs. Their predictions range from scalar scores to comparisons, localized defects, and tool-supported judgments. For scalar quality prediction, methods differ in their representations and training targets. CLIP-IQA uses image–text relations without quality-label fitting, while LIQE fine-tunes CLIP through joint quality, scene, and distortion prediction [19], [93]. QualiCLIP learns quality-sensitive alignment from rankings induced by synthetic distortions of pristine training images, without human opinion labels [94]. Q-Align learns text-defined quality levels and converts their probabilities into a score, whereas DeQA-Score learns quality-level distributions from human score

distributions [20], [95]. LL-Score assesses degraded–restored image pairs for quality and hallucination cues without a clean counterpart [174].

Comparisons and descriptions express relative quality and specific defects. Co-Instruct answers multi-image quality questions and explains relative differences [96]. Compare2Score compares a test image with quality anchors and converts preference probabilities into a continuous score. The anchors need not depict a clean version of the test scene [97]. Q-Instruct converts human low-level descriptions into instruction data for visual question answering [98]. DepictQA and DepictQA-Wild support descriptive assessment and comparison, with separate full-reference and no-reference modes and differing training resources [57], [99].

Spatial assessment associates quality judgments with particular image regions. Q-Ground predicts text-conditioned distortion masks [100]. SEAGULL assesses mask-specified regions using local and global representations. Its synthetic supervision uses full-reference quality labels, while assessment at inference is no-reference [101]. Grounding-IQA associates quality descriptions with locations and answers questions about local regions [102]. IPAD-CLIP adapts text and visual representations for local perceptual-artifact classification and localization [103]. SR-Ground trains models to localize artifacts in super-resolved images using low-resolution references, then uses their frozen grounding predictions to condition OSEDiff fine-tuning [175].

Reasoning-oriented assessors differ in how they learn quality judgments and use reasoning at inference. Q-Insight uses reinforcement learning for score regression and degradation perception, while VisualQuality-R1 learns relative quality through ranking rewards [104], [105]. Q-Ponder combines supervised reasoning initialization with subsequent reinforcement learning, and Q-Hawkeye uses rollout-score uncertainty to reweight training updates [176], [177]. RALI instead transfers reasoning-derived representations into an image scorer, without an LLM or text generation during scoring [107].

Tool-supported assessment gathers additional evidence through specialized analyses. IQA-T1 gathers tool-generated visual evidence, including noise residuals, gradient statistics, and frequency spectra, for quality reasoning [56]. AgenticIQA selects compatible quality tools, combines their results, and can revise its assessment plan when evidence is insufficient [106]. It operates in full-reference or no-reference mode according to the supplied inputs. Such assessment outputs also provide evidence for the runtime restoration feedback discussed in Section IV-C.

III. Restoration-Oriented Foundation Models

This section reviews learning strategies toward restoration-oriented foundation models, together with unified restorers that provide relevant design precedents. We examine pretraining, unified restoration modeling, and transfer and adaptation. These complementary strategies concern representation learning, task coverage, and extension to new tasks or domains. Table II summarizes representative methods.

A. Pretraining for Image Restoration

Pretraining learns model parameters or components for subsequent restoration training through multiple restoration tasks, masked reconstruction, or degradation-aware learning.

1) Multi-Task Pretraining: Multi-task pretraining learns representations and image priors from multiple restoration tasks to provide a starting point for subsequent restoration training. IPT pretrains a shared transformer with task-specific heads, tails, and embeddings on synthetic ImageNet pairs, using reconstruction and patch-level contrastive losses [21]. Task-specific fine-tuning then supports transfer across degradation strengths and blur. After $\times 2$ super-resolution (SR) fine-tuning, EDT compares peak signal-to-noise ratio (PSNR) gains from two pretraining settings. Reported gains are larger for pretraining on $\times 2$, $\times 3$, and $\times 4$ SR than for the tested mixture of $\times 2$ and $\times 3$ SR with Gaussian denoising at $\sigma = 15$, a difference associated with representation mismatch [194]. TAPE extracts prior queries from clean targets during pretraining and auxiliary pseudo-clean predictions during fine-tuning and inference [22]. Diffusion-inspired training uses generative pretraining before constrained reconstruction and generation fine-tuning [195]. Its unified setting orders tasks by matched diffusion timesteps and deploys the restoration network for trained-task and cross-dataset evaluations.

2) Masked Image Modeling: Masked image modeling provides a pretraining objective by reconstructing masked image content from visible regions [196]. MAEIP first pretrains its encoder through masked-pixel prediction, then pretrains the full autoencoder through whole-image reconstruction [197]. For restoration pretraining, predicting clean targets from masked degraded inputs also learns degradation removal. RAM predicts clean pixels from masked degraded inputs [23]. It uses mask attribute conductance along masked-to-unmasked inputs to rank layers, then fine-tunes the highest-ranked layers on unmasked degraded images. DyNet uses masked restoration during dynamic pretraining, randomly switching between shared-weight deep and shallow variants [198]. Module reuse determines their depth difference, and restoration fine-tuning follows pretraining. Masked reconstruction can also be combined with mixed-image separation during pretraining. CPNet uses this combination with parallel encoders and context fusion to learn representations for blind image decomposition [192].

3) Degradation-Aware Pretraining: Degradation-aware pretraining learns representations or conditioning components through tasks such as degradation transfer and classification, providing degradation-related information for subsequent image restoration. DegAE learns reference-image degradation transfer, then replaces its decoder for restoration fine-tuning [199]. DCPT alternates classification and reconstruction pretraining before restoration fine-tuning; its DC-guided cross-degradation transfer retains a frozen classifier [24]. A degradation prompter can instead remain fixed while different restorers are trained. ConStyle v2 pretrains a prompter on degraded images with auxiliary ImageNet-label classification, then freezes it to condition different restorers [188]. This method is evaluated across eight degradation types. A related staged generative approach connects prior pretraining

Table II. Representative restoration-oriented foundation models and learning designs.

Method	FM / Backbone	Role in restoration	Guidance information
IPT [21]	Transformer + task heads	Multi-task pretraining	Degraded image and task embedding
UniRestorer [178]	DA-CLIP + restoration experts	Expert selection and restoration	Degradation and granularity estimates
PROD + CoLoRA [179]	NAFNet or Restormer	Restoration adaptation	Degraded image
ProRes [180]	MAE ViT-L + decoder	Prompt-conditioned restoration	Image and degradation-aware visual prompt
Painter [181]	ViT-L	In-context prediction	Query and input-output example pair
GenLV [28], [182]	X-Restormer + prompt encoder	In-context restoration	Query and input-output example pair
Hidden-Shot [183]	Painter or MAE + CLIP	In-context prediction	Query, example pair, and task text
FoundIR [25]	Diffusion generalist + specialists	Degradation removal and refinement	Degraded image and generalist output
FoundIR-v2 [184]	SDXL soft-MoE	Task-adaptive restoration	LQ features and noisy target latents
Lumina-OmniLV [29]	Lumina-Next	Unified low-level generation	Degraded image and text instruction
LaVin-DiT [185]	Spatiotemporal VAE + DiT	In-context image and video generation	Query and input-target context pairs
UARE [186]	BAGEL dual experts	Quality-guided image generation	Image, instruction, and quality-analysis tokens
SLAIR [187]	Structured VAE + latent flow	Latent-space restoration	Degraded-image latent
DCPT [24]	Restorer + ResNet18	Encoder pretraining	Degraded image and degradation labels
ConStyle v2 [188]	ConStyle CNN + restorer	Clean visual prompt extraction	Degraded image and clean visual prompts
BDG [189]	ImageNet U-Net or SD2	Degradation-aware pretraining	LQ image or latent
AdaIR [190]	Restormer + adapters	Restoration adaptation	Degraded image
RestoreMore [191]	Restorer + residual experts	Continual restoration	Degraded-image features for routing
CPNet [192]	Parallel transformer encoders	Context-aware pretraining	Masked reconstruction and mixed-image separation
Filter-level RwF [193]	Restorer + filter generators	Filter-level continual restoration	Filter attribution and task prototypes

with degradation-aware learning. BDG first learns a generation prior and then aligns degradation-discriminative and diffusion features in a bridging stage before restoration fine-tuning [189].

B. Unified Restoration Modeling

Unified restoration models learn to handle multiple tasks through shared parameters, visual examples, or generative components.

1) Multi-Task Joint Training: Multi-task joint training optimizes a restoration model across multiple tasks. Task-specific conditioning or expert selection can help handle different degradations. The early all-weather model uses task-specific encoders and a shared decoder [200], whereas TransWeather uses learned weather queries in one encoder-decoder [201]. IDR integrates task-specific prior hubs into a shared embedding hub queried by degraded features through cross-attention for correction [202]. AirNet learns contrastive degradation representations [203], PromptIR combines input-dependent prompts [204], and OneRestore uses scene descriptions [205]. UniRestorer selects experts at different granularities using degradation estimates and uncertainty [178]. These designs govern input-dependent responses, while the training schedule determines how tasks are learned together. MiO adds tasks progressively while retaining earlier training data [206].

2) Visual In-Context Learning: Visual in-context learning uses example input-output pairs to specify the desired transformation and restore a query image, with model parameters fixed during prediction. Painter learns masked outputs in a shared image format [181]. During training, PromptGIP masks the answer images for both example and query pairs [27]. GenLV injects separately encoded visual prompts through cross-attention [28], while Hidden-Shot adds task text and implicit prompts to one-shot examples [183]. LaVin-DiT combines a spatiotemporal VAE with an example-pair-conditioned diffusion transformer for image and video prediction using conditional flow matching [185].

Restoration quality and performance on new tasks depend on both the selected example pairs and the tasks covered during training or prior adaptation. PromptGIP analyzes the best of 20 prompts [27]. GenLV expands from 30 to 101 trained tasks, evaluating fixed-parameter inference, five-pair backbone fine-tuning, and separate task-specific fine-tuning [182]. Hidden-Shot fixes parameters during final prediction, but 3C4U includes prior full-shot adaptation whereas 3C7U retains three-task training [183]. Claims of unseen-task transfer without adaptation require excluding the target task from both training and pre-test adaptation. Painter segmentation results do not establish unseen-task restoration [181]. PromptGIP reports mixed-degradation and colorization failures [27], and GenLV reports unseen-task limitations [28]. LaVin-DiT reports good performance on related unseen tasks but difficulty with substantially different task definitions [185].

3) Multi-Task Generative Modeling: Multi-task generative models learn to restore images across different tasks through shared generative components or combinations of generalist and specialist models. Jointly training one checkpoint differs from independently training the same architecture for different tasks. Palette concatenates the source image with the noisy target and also studies one jointly trained image-to-image checkpoint [207]. FoundIR trains a residual diffusion generalist and specialists on million-scale real pairs [25]. Training progresses from isolated to coupled degradations while retaining earlier data. The complete composition is evaluated across datasets. At inference, the fixed composition uses specialists to refine the generalist output. Results on coupled degradations represented in training describe task coverage rather than unseen-task performance. Across such mixtures, task sampling and paired-target quality also shape training. FoundIR-v2 uses reference-set performance changes for sampling and multimodal assessment for target filtering [184]. Its SDXL-based soft attention MoE scheduler conditions on low-quality features and noisy target latents.

Joint generative training can combine restoration with other low-level vision tasks, with text instructions specifying the de-

sired transformation. Lumina-OmniLV jointly trains a Lumina-Next-initialized flow backbone and condition adapters on approximately 40 million instances across over 100 subtasks, including 45 restoration subtasks [29]. Training progresses from single-image to in-context tasks and then to higher resolution. Visual examples are used for stylization.

Quality analysis can guide restoration during inference, while degradation classification can help organize the latent representation during training. UARE first trains its restoration expert, then jointly tunes assessment and restoration experts [186]. Quality-analysis tokens condition restoration at inference, supporting assessment, restoration, and enhancement with real-SR evaluation. SLAIR instead structures VAE latents through clean-versus-degraded and degradation-type classification during training, then learns a paired latent velocity field that is integrated from degraded latents at inference [187].

C. Transfer and Adaptation

1) Transfer to New Tasks and Domains: Transfer can involve new tasks, data distributions, or both, with model parameters either fixed or adapted. MPerceiver reports direct transfer to under-display-camera and underwater restoration, and few-shot adaptation for other tasks [147]. Its generative components are described in Section II-B. GRIDS groups tasks by degradation statistics and selects trained group restorers according to input-descriptor distance, without test-time parameter updates [208].

For parameter-efficient adaptation, CoLoRA adapts PROD through low-rank, bias, and normalization updates [179]. PROD is pretrained with randomly ordered and potentially repeated degradations. ProRes uses an MAE-initialized ViT backbone, a decoder, and image-shaped input prompts [180]. It tunes the prompts for new data and tasks while keeping the backbone fixed. The adapter-based AdaIR trains adapters in a frozen Restormer [190], while AdaptIR trains heterogeneous experts in frozen IPT or EDT backbones [26].

Without paired target-domain data, DDTNet transfers target degradations to source clean images [209]. Degradation-consistent adaptation instead uses teacher pseudo-targets and a re-degradation generator learned from test observations [210]. It preserves parameters identified as important by Fisher information and updates the teacher through an exponential moving average.

2) Continual Image Restoration: Continual image restoration learns tasks sequentially while seeking to preserve performance on previously learned tasks. Revisiting earlier training samples helps maintain previously acquired restoration capabilities. Knowledge replay combines stored images from earlier tasks with distillation [211]. Review Learning also revisits earlier samples during sequential learning. It orders restoration tasks by degradation complexity, measured by the entropy difference between clean and degraded images, and reviews retained challenging samples when learning subsequent tasks [212].

Task-specific parameters provide another way to accommodate successive restoration tasks. MINI-Conv generates task-specific convolution parameters from lightweight embeddings [213]. Restoring Without Forgetting learns low-rank paths from paired data within a frozen denoiser and retains previously learned paths [214]. Prototype-based routing selects these paths without requiring degradation labels at test time. Its evaluation covers five sequential domains and cross-dataset transfer. Task-specific adaptation can also be localized to individual filters. Feng et al. [193] use parameter-space integrated gradients to identify degradation-critical filters, generate task-specific filters from a filter bank through compact low-rank transformations, and replace the selected filters while retaining the remaining network parameters. At the expert level, RestoreMore freezes earlier components and trains new residual experts and a router [191]. It sparsely combines experts to extend restoration from deblurring to deraining, desnowing, dehazing, and denoising.

IV. Agentic Image Restoration

Agentic restoration addresses how to organize these capabilities for a particular image and goal. Controllers choose operations and their order, select and configure models, and may assess intermediate results to revise actions or stop. Foundation models can serve as controllers, assessors, or generative executors. We organize these decisions by function rather than executor type. These functions can coexist within a system. Table III summarizes representative methods.

A. Task Planning

Task planning determines which restoration operations to perform and in what order. Each operation can alter the remaining degradations or introduce artifacts that affect subsequent restoration. AgenticIR uses GPT-4 to schedule tasks, guided by summaries of offline trials [30]. A LoRA-adapted DepictQA model assesses results, guiding decisions to retain an output, roll back an operation, or revise the plan. MAIR uses hierarchical LLMs, a tool registry, and prior experience to order stages addressing compression, imaging, and scene degradations [32]. Within each stage, expert agents test tools for the remaining degradations. DepictQA guides result acceptance and retries.

Other controllers predict a complete plan before execution. RestoreAgent learns from scored paths that include intermediate decisions, rollback, and stopping [33]. It uses complete plans by default, with stepwise replanning evaluated separately. JarvisIR uses quality-ranked plans for supervised and preference training [2]. OPERA trains its planner with GRPO and updates restoration tools using losses on the final restored output [215]. It executes complete plans without intermediate replanning.

Planning can also be applied to refine the output of a task-specific restoration model. Derain-Agent uses a ResNet34-based network to select a refinement path from predefined tool sequences and predict spatially adaptive tool strengths for a preliminary derained image [223]. Offline path search provides supervision for path selection, while the strength maps are learned separately using a paired reconstruction loss. At inference, the selected sequence is executed without iterative candidate search.

Table III. Representative restoration agents and related control mechanisms.

Method	FM / Backbone	Role in restoration	Guidance information
AgenticIR [30]	GPT-4 + DepictQA + tools	Scheduling and quality assessment	Degradation reports and trial summaries
Q-Agent [37]	Qwen2-VL-7B + tools	Perception and greedy selection	Images and no-reference quality scores
RestoreAgent [33]	LLaVA-Llama3-8B + tools	Task sequencing and model selection	Image, prompt, and execution history
Restore-R1 [38]	CLIP actor–critic	Sequential tool selection	Current image and action record
OPERA [215]	VisualQuality-R1 + tools	Restoration planning	Image and tool-list prompt
SEAR [216]	GPT-4o + DepictQA + tools	Trajectory retrieval and search	Degradation profile, memory, and IQA scores
SkillIR [35]	GPT-4o + DepictQA + tools	Tool selection and verification	Residual degradations, skills, and feedback
Causal-AgentIR [217]	GPT-4o + DepictQA + tools	Memory-guided planning and critique	Image, instruction, and memory graph
HarnessIR [36]	Gemini + image editors	Prompt composition and verification	Image, request, and tool evidence
FAPE-IR [218]	Qwen2.5-VL + FLUX	Restoration expert routing	Image, instruction, and frequency features
ExpertFusion [219]	VLM + diffusion LoRAs	Expert LoRA allocation	Image and user-intent features
RAR [220]	DepictQA + SD3.5	Latent assessment and restoration	Image latents and quality embeddings
CoR [221]	MobileNetV3 + restorer	Task selection and stopping	Current degradation probabilities
IAMAgent [222]	LLM manager + specialist agents	Multi-agent coordination	User commands, human feedback, and critique

B. Tool Selection

Tool selection determines which restoration model performs an operation. Configuration controls how selected models or their internal experts are combined or conditioned. Learned policies can select tools from the current image and action history. TIR-Agent selects both the restoration task and its model [34]. Its training combines demonstration fine-tuning with reinforcement learning using full-reference and no-reference signals. Restore-R1 selects actions through a visual actor-critic trained with DeQA-Score rewards, without runtime MLLM reflection or candidate search [38]. DiTTo uses a VLM to choose actions from the current image and tool history [224]. It first learns from simulated trajectories with action scores, then uses real trajectories for preference alignment. Adding experts requires further real alignment trajectories. These training signals are distinct from the runtime feedback discussed below.

Related executor-level mechanisms select or weight internal experts rather than schedule operations. They can configure a tool within an agent, but internal routing alone does not establish an agentic workflow. FAPE-IR trains executor-side frequency-aware LoRA-MoE modules, routed using a frozen VLM, frequency statistics, and instructions [218]. ExpertFusion uses an allocator trained on human preferences to predict diffusion task-LoRA weights from frozen VLM image and intent features [219].

C. Feedback and Refinement

Runtime feedback guides result acceptance, revision, and stopping, as also illustrated by AgenticIR and MAIR in Section IV-A. Q-Agent retains the candidate with the highest combined statistical and learned no-reference quality score at each round [37]. It stops when all detected targets are addressed or no candidate improves this score. Its MLLM perception module is trained through supervised fine-tuning. SkillIR verifies each tool output before accepting it as the current restoration state, then reassesses the remaining degradations to guide the next action [35]. 4KAgent uses LLM and VLM prompts and user profiles for planning and candidate assessment [225]. Its checks cover preferences, image quality, and face identity, with threshold-based criteria triggering revision.

User requests and feedback also guide execution. HybridAgent routes clear requests to a lightweight controller and ambiguous requests to an adapted multimodal controller [31]. Its trained feedback model uses restored images and tool history to decide whether processing should continue. Its tools include mixed-distortion LoRA restoration. Clarity ChatGPT supports language and regional interaction and switches restorers when users are dissatisfied [226]. IAMAgent incorporates human feedback into a multi-agent restoration framework, where a manager coordinates degradation perception, specialized restoration, and quality critique [222].

For generative executors, feedback can revise instructions or restoration conditions. HarnessIR uses frozen VLMs for diagnosis, instruction composition, and verification around a fixed generative executor [36]. OCR, face, depth, and segmentation tools provide content evidence. Verification checks corrections and protected content, using no-reference quality diagnostically rather than as a threshold. On failure, the controller revises instructions and retries from the original image for at most three rounds, retaining the better candidate. RAR uses adapters to map flow-restoration latents to assessor inputs and latent assessments to restoration conditions [220]. Joint training proceeds without decoding assessment text. Intermediate assessments update the restoration conditions, while comparisons of image quality before and after restoration determine whether processing continues. VLM-IMI tunes conditions around fixed SD2.1 and returns enhanced images to LLaVA for revised lighting instructions with user control [227].

D. Experience Retrieval

Controllers can retrieve experience from earlier restoration cases to supplement trained parameters and the execution history of the current image. Retrieved priorities and trajectories guide task ordering, individual actions guide the next tool call, and tool profiles summarize behavior across cases. EvoIR-Agent retrieves statistical and textual experience for tool priorities and order, with reflection, rollback, and memory updates [228]. Its fidelity and perceptual configurations use different assessment resources and incur data and assessment costs despite fixed controller weights.

SEAR evaluates a retrieved trajectory through a quality gate and falls back to tree search if it fails [216]. Its trajectory mem-

Table IV. Single-task generative $\times 4$ super-resolution results on RealSR and DRealSR. We report PSNR (↑), LPIPS (↓), MUSIQ (↑), and CLIP-IQA (↑). '–' marks an unavailable value. IFCSR uses $\gamma = 0.5$ with one sampling step per endpoint. # Steps denotes the number of sampling steps. # Params. includes the inference components. For methods with unclear configurations, we report the approximate parameter count of the restoration backbone.

Method	Publication	RealSR				DRealSR				# Steps	# Params	Key Features
		PSNR	LPIPS	MUSIQ	CLIP-IQA	PSNR	LPIPS	MUSIQ	CLIP-IQA			
ResShift [159]	NeurIPS 2023	26.31	0.3489	58.10	0.5450	28.45	0.4073	49.86	0.5259	15	180M	Residual-shifting diffusion
DiffBIR [45]	ECCV 2024	24.88	0.3567	64.66	0.6412	26.84	0.4446	60.68	0.6292	50	1680M	IRControlNet-conditioned SD
OSEDiff [49]	NeurIPS 2024	25.15	0.2921	69.09	0.6693	27.92	0.2968	64.65	0.6963	1	1770M	Variational score distillation
PASD [146]	ECCV 2024	25.22	0.3392	68.74	0.6502	27.48	0.3854	64.55	0.6714	20	2310M	Pixel-aware SD cross-attention
SeeSR [154]	CVPR 2024	25.33	0.2985	69.37	0.6594	28.26	0.3197	64.84	0.6672	50	2510M	Hard and soft prompts
SinSR [51]	CVPR 2024	26.30	0.3212	60.41	0.6204	28.41	0.3741	55.34	0.6367	1	180M	ResShift distillation
StableSR [10]	IJCV 2024	24.69	0.3091	65.42	0.6195	28.04	0.3354	58.50	0.6171	200	1560M	SD with time-aware encoder
PiSA-SR [164]	CVPR 2025	25.50	0.2672	70.15	0.6702	28.31	0.2960	66.11	0.6970	1	1300M	Pixel- and semantic-level LoRAs
AlloSR^2 [162]	ECCV 2026	24.97	0.2755	70.11	0.6750	27.95	0.2989	66.95	0.6883	1	$\approx$ 12000M	Allomorphic trajectory matching
DNF-SR [46]	CVPR 2026	24.97	0.3239	72.04	0.7257	28.14	0.3531	68.73	0.7559	1	$\approx$ 16871M	Dual-input FLUX-Kontext
FiDeSR [235]	CVPR 2026	26.02	0.2626	69.82	0.6896	28.90	0.2836	65.78	0.6974	1	1290M	SD latent residual refinement
FreqOrtho-SR [236]	ECCV 2026	26.27	0.2537	69.39	0.6630	29.08	0.2799	65.89	0.6879	1	1300M	Frequency-guided LoRAs + OGP
GDPO-SR [158]	CVPR 2026	25.48	0.2675	69.42	0.6760	28.18	0.2851	65.63	0.7020	1	1770M	Group DPO
GramSR [109]	ICPR 2026	26.81	0.2305	–	–	29.69	0.2603	–	–	–	$\approx$ 1049M	DINOv3 conditioning
IFCSR [165]	CVPR 2026	25.14	0.2866	69.68	0.6522	27.60	0.3278	66.57	0.7149	2×1	1300M	Endpoint image blending
ODTSR [166]	CVPR 2026	25.07	0.2398	68.29	–	28.14	0.2592	62.86	–	1	$\approx$ 20400M	Noise-hybrid visual stream
TADSR [151]	CVPR 2026	25.17	0.3168	71.18	0.7283	28.39	0.3235	67.02	0.7398	1	$\approx$ 1509M	Time-aware SD VAE encoder
TinySR [52]	CVPRF 2026	24.79	0.2806	69.78	0.7035	27.48	0.3116	65.36	0.7094	1	341M	SD3 DiT pruning

ory and search statistics can be revised or removed. SkillIR retrieves scene-aware skills that propose bounded actions [35]. Before committing to a candidate, an adapted verifier checks it for degradation reduction and content damage. Rejection preserves the current image, while accepted progress and resolved targets guide subsequent actions. Verified evidence updates skills after each case. PaAgent combines an adapted VLM for perception and task selection with behavioral-profile retrieval [229]. Its training uses GRPO, while runtime decisions use scores and descriptions. Causal-AgentIR uses multiple agents to selectively update, merge, and forget graph-memory evidence about tools, degradations, quality, cost, and preferences [217].

V. Comparative Evaluation

We summarize reported results under their respective benchmark settings and discuss the key technical components of selected representative methods. The tables provide more information, including key features, evaluation settings, performance, and parameter counts.

Evaluation Criteria. Image restoration is evaluated using reference-based, no-reference, and distribution-level metrics. PSNR measures pixel-level agreement, while SSIM compares structural similarity with a reference image [230]. LPIPS compares deep features of restored and reference images using a perceptually calibrated distance [231]. Without a reference image, MUSIQ predicts quality from multiscale image representations, CLIP-IQA uses image–text correspondence with quality prompts, and NIQE measures deviations from natural-image statistics [19], [232], [233]. At the distribution level, FID compares the means and covariances of Inception features extracted from restored and real image sets [234].

A. Single-Task Image Restoration

Super-Resolution. The real-world SR results in Table IV show that different approaches achieve different balances between reconstruction fidelity and perceptual quality. DNF-SR obtains higher no-reference quality scores, whereas PiSA-SR achieves better reference-based results on RealSR and DRealSR [46], [164]. FreqOrtho-SR improves both PSNR and LPIPS over PiSA-SR on these datasets, showing that the two aspects can also improve together [236]. The comparison in Table IX further shows that adding pixel-level adaptation to PiSA-SR improves reconstruction fidelity, highlighting the contribution of pixel-level supervision alongside semantic guidance [164].

For SR with known degradation operators, Table V(a) shows that DAPS improves both reconstruction and perceptual metrics over the compared fixed-prior methods on the two datasets. DAPS uses decoupled annealing to refine the image through posterior sampling at each noise level, illustrating how the inference procedure can affect restoration with a pretrained generative prior [140].

Denoising. Table VI suggests that the contribution of semantic guidance depends on the image content. The results of DINO-IR are consistent with pretrained features being particularly useful for reconstructing structured urban scenes [7]. CLIPDenoising also shows that a frozen pretrained encoder can support restoration at unseen noise levels, although the same benefit is not observed at the training noise level [238]. Together, these findings identify structure preservation and robustness to changes in noise as distinct contributions of pretrained guidance.

Deblurring. For motion deblurring, Table VII summarizes the results on GoPro. The matched comparison in Table IX further shows that incorporating SAM-derived region priors into NAFNet yields a substantial improvement on ReLoBlur while leaving GoPro performance nearly unchanged [113]. This result suggests that region-level structural guidance is particularly beneficial for cross-dataset generalization in this

Table V. Image restoration results on FFHQ256 and ImageNet256 with known degradation operators and noisy measurements. The degradation settings follow DAPS [140] and comprise (a) non-blind $\times 4$ SR, (b) non-blind Gaussian deblurring, and (c) box or random inpainting. We report PSNR (↑) and LPIPS (↓). '–' denotes an unavailable value. # Params. includes latent autoencoders where applicable.

Method	Publication	(a) SR		(b) Deblurring		(c) Box		(c) Random		# Params	Key Features
		PSNR	LPIPS	PSNR	LPIPS	PSNR	LPIPS	PSNR	LPIPS		
FFHQ256											
DDRM [16]	NeurIPS 2022	26.58	0.2820	24.93	0.2390	22.26	0.2070	–	–	–	SVD conditioning
DDNM [133]	ICLR 2023	28.03	0.1970	28.20	0.2160	24.47	0.2350	29.91	0.1210	–	Range-space correction
DPS [134]	ICLR 2023	25.86	0.2690	25.87	0.2190	22.51	0.2090	25.46	0.2030	94M	Approximate likelihood guidance
DiffPIR [138]	CVPRW 2023	26.64	0.2600	27.36	0.2360	–	–	–	–	94M	HQS-guided diffusion sampling
PSLD [135]	NeurIPS 2023	24.35	0.2870	23.27	0.3160	24.22	0.1580	30.31	0.2210	329M	Latent gluing guidance
DAPS [140]	CVPR 2025	29.07	0.1770	29.19	0.1650	24.07	0.1330	31.12	0.0980	94M	Decoupled annealing
LatentDAPS [140]	CVPR 2025	27.48	0.1820	27.93	0.2340	23.99	0.1940	30.71	0.1410	329M	Latent decoupled annealing
DCDP [139]	ICCP 2026	28.66	0.1780	27.50	0.3040	23.89	0.1630	30.69	0.1420	94M	Decoupled data consistency and purification
ImageNet256											
DDRM [16]	NeurIPS 2022	22.62	0.3240	21.26	0.4430	18.63	0.2540	–	–	553M	SVD conditioning
DDNM [133]	ICLR 2023	23.96	0.4750	28.06	0.2780	21.64	0.3190	31.16	0.1910	553M	Range-space correction
DPS [134]	ICLR 2023	21.13	0.3610	20.31	0.3970	18.94	0.2570	23.52	0.2970	553M	Approximate likelihood guidance
DiffPIR [138]	CVPRW 2023	23.18	0.3710	22.80	0.3550	–	–	–	–	553M	HQS-guided diffusion sampling
PSLD [135]	NeurIPS 2023	25.42	0.3600	25.86	0.3900	20.10	0.4650	31.30	0.3370	$\approx$ 1066M	Latent gluing guidance
DAPS [140]	CVPR 2025	25.89	0.2760	26.15	0.2530	21.43	0.2140	28.44	0.1350	553M	Decoupled annealing
LatentDAPS [140]	CVPR 2025	25.06	0.2760	25.05	0.3450	17.19	0.3400	27.59	0.1640	457M	Latent decoupled annealing

Table VI. Single-task Gaussian denoising results on BSD68, CBSD68, and Urban100. We report PSNR (↑) and SSIM (↑). '–' denotes an unavailable value. # Params. includes the inference components. For methods with unclear configurations, we report the approximate parameter count of the restoration backbone.

Method	Publication	Test set	$\sigma = 15$		$\sigma = 25$		$\sigma = 50$		# Params	Key Features
			PSNR	SSIM	PSNR	SSIM	PSNR	SSIM		
PromptIR [204]	NeurIPS 2023	BSD68	34.34	.938	31.71	.897	28.49	.813	36M	Input-conditioned prompts
DINO-IR [7]	arXiv 2023	BSD68	34.34	.943	31.72	.903	28.46	.821	$\approx$ 66M	Multilayer DINOv2 fusion
VLU-Net [237]	CVPR 2025	CBSD68	34.35	–	31.72	–	28.46	–	$\approx$ 123M	CLIP-guided unfolding
Perceive-IR [169]	TIP 2026	CBSD68	34.38	–	31.74	–	28.53	–	$\approx$ 129M	CLIP loss, DINOv2 guidance
ClearAIR [118]	AAAI 2026	BSD68	34.39	–	31.73	–	28.52	–	$\approx$ 31M	DeQA and SAM guidance
QuReC [120]	ACM MM 2026	CBSD68	34.41	.936	31.78	.895	28.56	.813	30M	CLIP prototype matching
PromptIR [204]	NeurIPS 2023	Urban100	34.77	.952	32.49	.929	29.39	.881	36M	Input-conditioned prompts
DINO-IR [7]	arXiv 2023	Urban100	34.91	.961	32.68	.941	29.60	.896	$\approx$ 66M	Multilayer DINOv2 fusion
VLU-Net [237]	CVPR 2025	Urban100	34.92	–	32.71	–	29.61	–	$\approx$ 123M	CLIP-guided unfolding
Perceive-IR [169]	TIP 2026	Urban100	34.86	–	32.55	–	29.42	–	$\approx$ 129M	CLIP loss, DINOv2 guidance
ClearAIR [118]	AAAI 2026	Urban100	34.84	–	32.54	–	29.41	–	$\approx$ 31M	DeQA and SAM guidance
QuReC [120]	ACM MM 2026	Urban100	35.08	.952	32.91	.931	29.93	.888	30M	CLIP prototype matching

setting.

For deblurring with a known Gaussian kernel, Table V(b) compares methods that use pretrained generative priors. DAPS employs decoupled annealing to incorporate measurement information during posterior sampling [140]. Compared with DDNM, it achieves lower LPIPS on both datasets, while the PSNR ranking differs between FFHQ256 and ImageNet256. These results show that the relative performance of the sampling methods varies with the data distribution and evaluation criterion.

Evidence Across Different Settings. The matched comparisons in Table IX show that pretrained knowledge can support restoration through different forms of training supervision. DegAE pretraining improves dehazing with both SwinIR and Restormer, although the gains vary across backbones [199]. For deraining, SAM-based distillation and diffusion-based supervision improve the respective restoration networks, while the reward ablation in RGSUD indicates the benefit of degradation-aware quality assessment under its unpaired training setting [18], [40], [170].

Pretrained models and reference images can also provide guidance during restoration. DA-CLIP conditioning improves low-light restoration with NAFNet, while PFStorer uses reference-based personalization to improve face restoration over its non-personalized base model [8], [157]. Together, these comparisons illustrate the benefits of introducing pretrained knowledge through training supervision or inference-time guidance across different tasks.

For inpainting with known degradation operators, Table V(c) compares different ways of using pretrained generative priors. DAPS achieves lower LPIPS than DDNM across the reported settings, whereas the PSNR ranking varies with the dataset and mask type [133], [140]. This comparison shows that improvements in reference-based perceptual similarity do not always coincide with higher pixel-level fidelity.

Table VII. Single-task motion deblurring results on GoPro. We report PSNR (↑) and SSIM (↑). CSformer uses MAEIP initialization. FideDiff-S uses the original GoPro training pairs. SAM-Deblur includes its restoration network and SAM ViT-H mask preprocessing. # Params. includes the inference components. For methods with unclear configurations, we report the approximate parameter count of the restoration backbone.

Method	PSNR	SSIM	# Params
CSformer [197]	33.36	.965	–
IR-SDE [153]	30.70	.901	34M
DA-CLIP [8]	30.88	.903	≈ 88M
Restormer [239]	32.92	.961	26M
SAM-Deblur [113]	32.83	.960	≈ 658M
DCPT-Restormer [24]	33.12	.962	26M
FideDiff-S [160]	28.56	.911	≈ 1300M

Publication and Key Features. **CSformer** (arXiv 2023), Masked pretraining. **IR-SDE** (ICML 2023), Mean-reverting SDE. **DA-CLIP** (ICLR 2024), CLIP content conditioning. **Restormer** (CVPR 2022), Transposed channel attention. **SAM-Deblur** (ICASSP 2024), SAM mask average pooling. **DCPT-Restormer** (ICLR 2025), Degradation classification pretraining. **FideDiff-S** (ICLR 2026), Blur-trajectory consistency.

Table VIII. Single-task deraining results on Rain100L. We report PSNR (↑) and SSIM (↑). *Note.* The metrics of IPT and EDT-B are calculated on the Y-channel of YCbCr. CSformer uses MAEIP initialization. IPT, EDT-B, and CSformer are fine-tuned on the target task. All other methods use single-task training. # Params. includes the inference components. For methods with unclear configurations, we report the approximate parameter count of the restoration backbone.

Method	PSNR	SSIM	# Params
IPT [21]	41.62	.988	114M
EDT-B [194]	42.14	.990	12M
CSformer [197]	39.49	.978	–
PromptIR [204]	37.04	.979	36M
DINO-IR [7]	37.12	.980	≈ 66M
VLU-Net [237]	38.60	.984	≈ 123M
Perceive-IR [169]	38.41	.984	≈ 129M
ClearAIR [118]	38.52	.985	≈ 31M
QuReC [120]	39.14	.986	30M

Publication and Key Features. **IPT** (CVPR 2021), Multi-task ImageNet pretraining. **EDT-B** (IJCAI 2023), Related-task pretraining. **CSformer** (arXiv 2023), Masked pretraining. **PromptIR** (NeurIPS 2023), Input-conditioned prompts. **DINO-IR** (arXiv 2023), Multilayer DINOv2 fusion. **VLU-Net** (CVPR 2025), CLIP-guided unfolding. **Perceive-IR** (TIP 2026), CLIP loss, DINOv2 guidance. **ClearAIR** (AAAI 2026), DeQA and SAM guidance. **QuReC** (ACM MM 2026), CLIP prototype matching.

Table IX. Single-task restoration results with benefits from foundation models or pre-training. Rows compare matched baseline-enhanced pairs under the same evaluation setting. We report PSNR (↑) and corresponding gains. EDT-B ×2 SR, ART ×4 SR, and PTG-RM deraining results use Y-channel PSNR. PTG-RM is trained on Rain13K. RGSUD replaces the CLIP-IQA reward with DACLIP-IQA while keeping the unpaired training setup unchanged.

Method	Task	Test dataset and setting	Base method	Publication	Key Features (Contribution)	Baseline PSNR	Enhanced PSNR	Gain PSNR
EDT-B [194]	Super-resolution	Urban100, bicubic ×2	EDT-B	IJCAI 2023	Multi-related-task pre-training	33.80	34.27	+0.47
SAM prior tuning [39]	Super-resolution	Urban100, ×4	ART	arXiv 2023	SAM prior tuning (SPT) unit	27.77	28.17	+0.40
TAPE [22]	Super-resolution	DIV2K, resized transfer	TAPE-Net (scratch)	ECCV 2022	Task-agnostic pre-training	31.25	31.66	+0.41
PiSA-SR [164]	Super-resolution	RealSR	PiSA-SR V2	CVPR 2025	Adding pixel-level LoRA	24.13	25.50	+1.37
SAM prior tuning [39]	Denoising	BSD68, RGB $\sigma = 25$	ART	arXiv 2023	SAM prior tuning (SPT) unit	31.84	31.92	+0.09
EDT-B [194]	Denoising	CBSD68, RGB $\sigma = 25$	EDT-B	IJCAI 2023	Multi-related-task pre-training	31.73	31.76	+0.03
TAPE [22]	Denoising	SIDD, resized 256^2	TAPE-Net (scratch)	ECCV 2022	Task-agnostic pre-training	37.41	37.90	+0.49
CLIPDenoising [238]	Denoising	Kodak24, $\sigma = 50$	RN50, random encoder	CVPR 2024	Frozen CLIP encoder, no PFA	20.87	26.87	+6.00
SAM distillation [40]	Denoising	SenseNoise	Uformer	CVPR 2024	Semantic priors distillation	35.14	35.25	+0.11
DINO-IR V5 [7]	Deraining	Rain100L	DINO-IR V0	arXiv 2023	Multi-scale DINOv2 fusion	36.74	37.07	+0.33
MAEIP [197]	Deraining	Rain100L	CSformer-T	arXiv 2023	Two-stage pre-training variant	37.03	37.43	+0.40
SAM distillation [40]	Deraining	Rain200L	RCDNet	CVPR 2024	Semantic priors distillation	39.04	40.26	+1.22
DiffLoss [18]	Deraining	Rain100H, Rain13K-trained	EfDeRain	ACCV 2024	DiffLoss auxiliary loss	23.41	24.54	+1.13
PTG-RM [115]	Deraining	Rain100L	Restormer	CVPR 2024	PTG-RM with CLIP priors	38.99	39.27	+0.28
RGSUD [170]	Deraining	Rain100L, unpaired	RGSUD (CLIP-IQA)	CVPR 2026	CLIP-IQA to DACLIP-IQA	33.67	34.41	+0.74
DCPT [24]	Deblurring	GoPro	Restormer	ICLR 2025	Degradation classification pre-training	32.92	33.12	+0.20
SAM-Deblur [113]	Deblurring	GoPro	NAFNet32	ICASSP 2024	Mask average pooling + dropout	32.85	32.83	-0.02
SAM-Deblur [113]	Deblurring	ReLoBlur, GoPro-trained	NAFNet32	ICASSP 2024	Mask average pooling + dropout	25.26	32.29	+7.03
DegAE [199]	Dehazing	SOTS-Indoor	SwinIR	CVPR 2023	DegAE pretraining	29.83	36.71	+6.88
DegAE [199]	Dehazing	SOTS-Indoor	Restormer	CVPR 2023	DegAE pretraining	39.01	39.39	+0.38
DiffLoss [18]	Dehazing	Dense-Haze	TaylorFormer	ACCV 2024	DiffLoss auxiliary loss	15.02	15.19	+0.17
DiffLoss [18]	Low-light	LOL	IAT	ACCV 2024	DiffLoss auxiliary loss	19.89	20.11	+0.22
DA-CLIP [8]	Low-light	LOL	NAFNet	ICLR 2024	DA-CLIP embedding conditioning	23.09	23.72	+0.63
PFStorer [157]	Face restoration	Celeb-Ref, heavy	Face-tuned StableSR	CVPR 2024	Reference-based personalization	22.15	22.62	+0.47

B. Universal Image Restoration

Three-Task Setting. Table X shows that improvements in joint restoration are not uniform across tasks. In the original DCPT comparison, degradation-classification pretraining brings more pronounced gains in dehazing and deraining than in Gaussian denoising [24]. This supports degradation-classification pretraining for weather restoration under the reported settings, while showing that its contribution remains task-dependent.

Five-Task Setting. For the broader task set in Table XI, pretraining and expert specialization provide complementary evidence. DCPT shows that degradation-aware initialization also benefits deblurring and low-light enhancement [24]. The effectiveness of expert specialization depends on how experts are selected. Separate ablations in UniRestorer show that finer degradation groups favor in-distribution restoration, while coarser groups can better tolerate distribution shifts [178]. Together, these findings favor learning task-sensitive representations and allowing models to select appropriate specialization levels. Finer specialization should not be assumed to be consistently preferable.

Composite Degradations Setting. For composite restoration

Table X. Universal image restoration results under the three-task setting. We report PSNR (↑) and SSIM (↑). '–' denotes an unavailable value. For most works, # Params. includes the inference components. For methods with unclear configurations, we report the approximate parameter count of the restoration backbone.

Method	Publication	Dehazing SOTS		Deraining Rain100L		Denoising on CBSD68 $\sigma = 15$		$\sigma = 25$		$\sigma = 50$		Average		# Params	Key Features
		PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM		
DL [240]	TPAMI 2019	26.92	.931	32.62	.931	33.05	.914	30.41	.861	26.90	.740	29.98	.875	2M	Conditional weight generation
TKMANet [241]	CVPR 2022	30.41	.973	34.94	.972	33.02	.924	30.31	.820	23.80	.556	30.50	.849	29M	Two-stage knowledge learning
AirNet [203]	CVPR 2022	27.94	.962	34.90	.968	33.92	.933	31.26	.888	28.00	.797	31.20	.910	9M	Contrastive degradation cues
ADFNet [242]	AAAI 2023	28.13	.961	34.24	.965	33.76	.929	30.83	.871	27.75	.793	30.94	.904	8M	Dynamic filtering
DehazeFormer [243]	TIP 2023	29.58	.970	35.37	.969	33.01	.914	30.14	.858	27.37	.779	31.09	.898	25M	RescaleNorm, window attention
DRSformer [244]	CVPR 2023	29.02	.968	35.89	.970	33.28	.921	30.55	.862	27.58	.786	31.26	.901	34M	Sparse attention
FSNet [245]	TPAMI 2023	29.14	.968	35.61	.969	33.81	.930	30.84	.872	27.69	.792	31.42	.906	13M	Frequency selection
PIP-Restormer [246]	arXiv 2023	32.09	.981	38.29	.984	34.24	.936	31.60	.893	28.35	.806	32.91	.920	27M	Prompt-in-prompt learning
IDR [202]	CVPR 2023	29.87	.970	36.03	.971	33.89	.931	31.32	.884	28.04	.798	31.83	.911	15M	Ingredients-oriented prior hub
PromptIR [204]	NeurIPS 2023	30.58	.974	36.37	.972	33.98	.933	31.31	.888	28.06	.799	32.06	.913	36M	Input-conditioned prompts
GridFormer [247]	IJCV 2024	30.37	.970	37.15	.972	33.93	.931	31.37	.887	28.11	.801	32.19	.912	34M	Residual dense grid transformer
ProRes [180]	arXiv 2023	28.38	.938	33.68	.954	32.10	.907	30.18	.863	27.58	.779	30.38	.888	371M	Image-shaped input prompts
MambaIR [248]	ECCV 2024	29.57	.970	35.42	.969	33.88	.931	30.95	.874	27.74	.793	31.51	.907	27M	Residual state-space blocks
NDR [249]	TIP 2024	28.64	.962	35.42	.969	34.01	.932	31.36	.887	28.10	.798	31.51	.910	28M	Degradation dictionary queries
Art-PromptIR [250]	ACM MM 2024	30.83	.979	37.94	.982	34.06	.934	31.42	.891	28.14	.801	32.48	.917	36M	Active task reweighting
AnyIR (2024) [251]	arXiv 2024	31.38	.979	37.90	.981	33.95	.933	31.29	.889	28.03	.797	32.51	.916	6M	Local-global intertwining
DaAIR [252]	arXiv 2024	32.30	.981	37.10	.978	33.92	.930	31.26	.884	28.00	.792	32.52	.913	6M	Low-rank expert routing
MEASNet [253]	arXiv 2024	31.61	.981	39.00	.985	34.12	.935	31.46	.892	28.19	.803	32.88	.919	31M	Expert selection
U-WADN [254]	arXiv 2024	29.21	.971	35.36	.968	33.73	.931	31.14	.886	27.92	.793	31.47	.910	6M	Adaptive network width
Shi et al. [255]	arXiv 2024	29.20	.972	37.50	.980	34.59	.941	31.83	.900	28.46	.814	32.32	.921	32M	Frequency-aware transformers
HAIR [256]	arXiv 2024	30.98	.979	38.59	.983	34.16	.935	31.51	.892	28.24	.803	32.70	.918	29M	Classifier-guided hypernetwork
LoRA-IR [117]	arXiv 2024	30.68	.961	37.75	.979	34.06	.935	31.42	.891	28.18	.803	32.42	.914	–	CLIP-routed low-rank experts
ABAIR-OH [257]	arXiv 2024	33.53	.984	38.69	.982	34.18	.935	31.38	.890	28.25	.804	33.21	.919	$\approx$ 41M	One-hot LoRA routing
MTAIR [258]	arXiv 2024	31.34	.983	39.15	.984	34.14	.936	31.50	.893	28.24	.805	32.87	.920	–	Spatial and channel prompts
InstructIR [3]	ECCV 2024	30.22	.959	37.98	.978	34.15	.933	31.52	.890	28.30	.804	32.43	.913	$\approx$ 33M	BGE instruction guidance
DA-CLIP (IR-SDE) [8]	ICLR 2024	26.83	.962	35.85	.972	31.43	.889	25.05	.605	18.33	.308	27.50	.747	174M	CLIP content-degradation cues
DyNet-S [198]	arXiv 2024	30.80	.980	38.02	.982	34.07	.935	31.41	.892	28.15	.802	32.49	.918	16M	Shared-weight depth variants
DyNet-L [198]	arXiv 2024	31.34	.980	38.69	.983	34.09	.935	31.43	.892	28.18	.803	32.75	.919	16M	Shared-weight depth variants
UP-Restorer [259]	AAAI 2025	30.68	.977	36.74	.978	33.99	.933	31.33	.888	28.07	.799	32.16	.915	28M	Prompt-guided unrolling
TUR-PromptIR [260]	AAAI 2025	31.17	.978	38.57	.984	34.06	.932	31.40	.887	28.13	.797	32.67	.916	36M	Task uncertainty regularization
AdaIR (frequency) [261]	ICLR 2025	31.06	.980	38.64	.983	34.12	.935	31.45	.892	28.19	.802	32.69	.918	29M	Frequency modulation
DA-RCOT [262]	TPAMI 2025	31.26	.977	38.36	.983	33.98	.934	31.33	.890	28.10	.801	32.61	.917	50M	Residual-conditioned OT
MoCE-IR [263]	CVPR 2025	31.34	.979	38.57	.984	34.11	.932	31.45	.888	28.18	.800	32.73	.917	25M	Complexity experts
DCPT-PromptIR [24]	ICLR 2025	31.91	.981	38.43	.983	34.17	.933	31.53	.889	28.30	.802	32.87	.918	35M	Degradation classification (DCPT)
RamIR [264]	Appl. Intell. 2025	31.29	.977	38.16	.981	34.04	.931	31.61	.891	28.19	.801	32.66	.916	22M	Reasoning and action prompting
Cat-AIR [265]	arXiv 2025	31.49	.980	38.43	.983	34.11	.935	31.44	.892	28.14	.803	32.72	.919	$\approx$ 32M	Content-adaptive attention
DSwinIR [266]	arXiv 2025	31.86	.980	37.73	.983	34.12	.933	31.59	.890	28.31	.803	32.72	.918	24M	DSwin attention
CPL-PromptIR [267]	arXiv 2025	31.27	.980	38.77	.985	34.15	.933	31.50	.889	28.23	.800	32.78	.917	36M	Contrastive prompt learning
AnyIR (2025) [268]	arXiv 2025	31.75	.981	38.61	.984	34.12	.936	31.46	.893	28.20	.804	32.83	.920	6M	Local-global gating
MIRAGE [269]	arXiv 2025	31.86	.981	38.94	.985	34.12	.935	31.46	.891	28.19	.803	32.91	.919	10M	Manifold contrastive learning
BaryIR [270]	arXiv 2025	31.33	.980	38.95	.984	34.16	.935	31.54	.892	28.25	.802	32.85	.919	$\approx$ 54M	Optimal-transport barycenters
SIPL-PromptIR [271]	arXiv 2025	31.09	.977	38.43	.984	34.12	.933	31.48	.889	28.22	.800	32.67	.917	39M	Dictionary-guided refinement
TextPromptIR [123]	EAAI 2025	31.65	.978	38.41	.982	34.17	.936	31.52	.893	28.26	.805	32.80	.919	234M	BERT text guidance
VLU-Net [237]	CVPR 2025	30.71	.980	38.93	.984	34.13	.935	31.48	.892	28.23	.804	32.70	.919	$\approx$ 124M	CLIP-guided unfolding
DFPIR [272]	CVPR 2025	31.87	.980	38.65	.982	34.14	.935	31.47	.893	28.25	.806	32.88	.919	$\approx$ 95M	CLIP-guided feature perturbation
Perceive-IR [169]	TIP 2026	30.87	.975	38.29	.980	34.13	.934	31.53	.890	28.31	.804	32.63	.917	$\approx$ 129M	CLIP loss, DINOv2 guidance
ClearAIR [118]	AAAI 2026	31.08	.981	38.61	.984	34.18	.935	31.50	.891	28.31	.804	32.74	.919	$\approx$ 31M	DeQA and SAM 2 guidance
MoFE-MLLM [119]	arXiv 2026	32.13	.982	38.34	.983	34.23	.938	31.61	.896	28.39	.812	32.94	.922	$\approx$ 3792M	MLLM-guided frequency routing
QuReC [120]	ACM MM 2026	32.77	.985	38.80	.985	34.26	.934	31.60	.892	28.34	.806	33.15	.920	30M	CLIP prototype matching
MGN-AIR [121]	ACM MM 2026	33.21	.987	39.62	.986	–	–	–	–	–	–	33.17	.915	34M	Pixel-wise restoration matrix
SLER-IR [122]	arXiv 2026	33.00	.982	38.34	.983	34.25	.936	31.68	.897	28.45	.813	33.14	.922	–	CLIP-guided expert routing
UniRestorer [178]	ICLR 2026	36.44	.984	41.68	.996	–	–	–	–	–	–	34.48	.958	627M	Multi-granularity expert routing

in Table XII, both degradation types and their relative effects must be considered. AllRestorer shows that combining visual and textual descriptors improves reconstruction, while removing adaptive restoration weights produces more artifacts [280]. These results suggest that degradation descriptions and adaptive restoration weights play complementary roles. The descriptions indicate what needs correction, while the weights determine how strongly each correction is applied. Thus, effective composite restoration requires coordinated treatment of coexisting degradations, not merely broader coverage of individual tasks.

Other Settings. Beyond these universal restoration settings,

Table XI. Universal image restoration results under the five-task setting. We report PSNR (↑) and SSIM (↑). '–' denotes an unavailable value. DCPT denotes degradation classification pre-training. For most works, # Params. includes the inference components. For methods with unclear configurations, we report the approximate parameter count of the restoration backbone.

Method	Publication	Dehazing SOTS		Deraining Rain100L		Denoising BSD68		Deblurring GoPro		Low-light LOL		Average		# Params	Key Features
		PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM		
DL [240]	TPAMI 2019	20.54	.826	21.96	.762	23.09	.745	19.86	.672	19.83	.712	21.06	.743	2M	Conditional weight generation
MIRNet-v2 [273]	TPAMI 2022	24.03	.927	33.89	.954	30.97	.881	26.30	.799	21.52	.815	27.34	.875	6M	Multi-scale residual blocks
DGUNet [274]	CVPR 2022	24.78	.940	36.62	.971	31.10	.883	27.25	.837	21.87	.823	28.32	.891	17M	Generalized unfolding
TransWeather [201]	CVPR 2022	21.32	.885	29.43	.905	29.00	.841	25.12	.757	21.21	.792	25.22	.836	38M	Weather queries
TAPE [22]	ECCV 2022	22.16	.861	29.67	.904	30.18	.855	24.47	.763	18.97	.621	25.09	.801	1M	Pseudo-clean prior queries
AirNet [203]	CVPR 2022	21.04	.884	32.98	.951	30.91	.882	24.35	.781	18.18	.735	25.49	.847	9M	Contrastive degradation cues
ADFNet [242]	AAAI 2023	24.18	.928	32.97	.943	31.15	.882	25.79	.781	21.15	.823	27.05	.871	8M	Dynamic filtering
DehazeFormer [243]	TIP 2023	25.31	.937	33.68	.954	30.89	.880	25.93	.785	21.31	.819	27.42	.875	25M	RescaleNorm, window attention
DRSformer [244]	CVPR 2023	24.66	.931	33.45	.953	30.97	.881	25.56	.780	21.77	.821	27.28	.873	34M	Sparse attention
HI-Diff [275]	NeurIPS 2023	25.09	.935	33.26	.951	30.61	.878	26.48	.800	22.01	.870	27.49	.887	24M	Latent diffusion priors
Retinexformer [276]	ICCV 2023	24.81	.933	32.68	.940	30.84	.880	25.09	.779	22.76	.863	27.24	.879	2M	Illumination-guided attention
FSNet [245]	TPAMI 2023	25.53	.943	36.07	.968	31.33	.883	28.32	.869	22.29	.829	28.71	.898	13M	Frequency selection
IDR [202]	CVPR 2023	25.24	.943	35.63	.965	31.60	.887	27.87	.846	21.34	.826	28.34	.893	15M	Ingredients-oriented prior hub
PIP-Restormer [246]	arXiv 2023	32.11	.979	38.09	.983	30.94	.877	28.61	.861	24.06	.859	30.76	.912	27M	Prompt-in-prompt learning
PromptIR [204]	NeurIPS 2023	26.54	.949	36.37	.970	31.47	.886	28.71	.881	22.68	.832	29.15	.904	36M	Input-conditioned prompts
DASL-MPRNet [277]	arXiv 2023	25.82	.947	38.02	.980	31.57	.890	26.91	.823	20.96	.826	28.66	.893	15M	SVD-inspired operators
GridFormer [247]	IJCV 2024	26.79	.951	36.61	.971	31.45	.885	29.22	.884	22.59	.831	29.33	.904	34M	Residual dense grid transformer
MambaIR [248]	ECCV 2024	25.81	.944	36.55	.971	31.41	.884	28.61	.875	22.49	.832	28.97	.901	27M	Residual state-space blocks
Art-Transweather [250]	ACM MM 2024	29.93	.908	22.09	.891	29.43	.843	25.61	.776	21.99	.811	25.81	.846	38M	Active task reweighting
DaAIR [252]	arXiv 2024	31.97	.980	36.28	.975	31.07	.878	29.51	.890	22.38	.825	30.24	.910	6M	Low-rank expert routing
AnyIR-2024 [251]	arXiv 2024	29.84	.977	36.91	.977	31.15	.882	26.86	.822	23.50	.845	29.65	.901	6M	Local-global intertwining
MEASNet [253]	arXiv 2024	31.05	.980	38.32	.982	31.40	.888	29.41	.890	23.00	.845	30.64	.917	31M	Expert selection
HAIR [256]	arXiv 2024	30.62	.978	38.11	.981	31.49	.891	28.52	.874	23.12	.847	30.37	.914	29M	Classifier-guided hypernetwork
ABAIR [257]	arXiv 2024	33.46	.983	38.18	.983	31.38	.898	29.00	.878	24.20	.865	31.24	.921	$\approx$ 41M	Low-rank adapter routing
InstructIR [3]	ECCV 2024	27.10	.956	36.84	.973	31.40	.887	29.40	.886	23.00	.836	29.55	.908	$\approx$ 33M	BGE instruction guidance
TUR-TransWeather [260]	AAAI 2025	29.68	.966	33.09	.952	30.40	.869	26.63	.815	23.02	.838	28.56	.888	38M	Task uncertainty regularization
AdaIR-frequency [261]	ICLR 2025	30.53	.978	38.02	.981	31.35	.889	28.12	.858	23.00	.845	30.20	.910	29M	Frequency modulation
DA-RCOT [262]	TPAMI 2025	30.96	.975	37.87	.980	31.23	.888	28.68	.872	23.25	.836	30.40	.910	50M	Residual-conditioned OT
MoCE-IR [263]	CVPR 2025	30.48	.974	38.04	.982	31.34	.887	30.05	.899	23.00	.852	30.58	.919	25M	Complexity experts
DCPT-PromptIR [24]	ICLR 2025	30.72	.977	37.32	.978	31.32	.885	28.84	.877	23.35	.840	30.31	.911	35M	Degradation classification (DCPT)
RamIR [264]	Appl. Intell. 2025	31.09	.979	37.56	.979	31.44	.886	28.82	.878	22.02	.828	30.19	.910	22M	Reasoning and action prompting
D3Net [278]	arXiv 2025	32.57	.965	38.35	.973	31.73	.860	32.70	.851	26.49	.857	32.37	.901	38M	Dynamic decomposition
Cat-AIR [265]	arXiv 2025	30.88	.978	38.21	.982	31.38	.890	28.91	.876	23.46	.848	30.57	.915	$\approx$ 33M	Content-adaptive attention
DSwinIR [266]	arXiv 2025	30.09	.975	37.77	.982	31.34	.885	29.17	.879	22.64	.843	30.20	.913	24M	DSwin attention
CPL-PromptIR [267]	arXiv 2025	30.82	.978	38.20	.983	31.41	.887	28.72	.874	23.65	.855	30.56	.915	36M	Contrastive prompt learning
AnyIR-2025 [268]	arXiv 2025	31.41	.980	38.51	.983	31.37	.891	28.85	.883	23.11	.856	30.65	.919	6M	Local-global gating
MIRAGE [269]	arXiv 2025	31.45	.980	38.92	.985	31.41	.892	28.10	.858	23.59	.858	30.69	.915	10M	Manifold contrastive learning
BaryIR [270]	arXiv 2025	31.12	.976	38.05	.981	31.43	.891	29.30	.888	23.38	.852	30.66	.918	$\approx$ 54M	Optimal-transport barycenters
VLU-Net [237]	CVPR 2025	30.84	.980	38.54	.982	31.43	.891	27.46	.840	22.29	.833	30.11	.905	$\approx$ 124M	CLIP-guided unfolding
DFPIR [272]	CVPR 2025	31.64	.979	37.62	.978	31.29	.889	28.82	.873	23.82	.843	30.64	.912	$\approx$ 95M	CLIP-guided feature perturbation
CyclicPrompt [279]	TIP 2025	33.02	.983	37.03	.989	31.33	.941	29.42	.925	21.79	.837	30.52	.935	$\approx$ 429M	Cyclic CLIP prompting
DCPT-SwinIR [24]	ICLR 2025	28.67	.973	35.70	.974	31.16	.882	26.42	.807	20.38	.836	28.47	.894	11M	Degradation classification (DCPT)
DCPT-NAFNet [24]	ICLR 2025	29.47	.971	35.68	.973	31.31	.886	29.22	.883	23.52	.855	29.84	.914	68M	Degradation classification (DCPT)
DCPT-Restormer [24]	ICLR 2025	29.86	.973	36.68	.975	31.46	.888	28.95	.879	23.26	.842	30.04	.911	26M	Degradation classification (DCPT)
Perceive-IR [169]	TIP 2026	28.19	.964	37.25	.977	31.44	.887	29.46	.886	22.88	.833	29.84	.909	$\approx$ 129M	CLIP loss, DINOv2 guidance
ClearAIR [118]	AAAI 2026	30.12	.978	38.20	.982	31.53	.888	29.67	.887	22.83	.846	30.47	.916	$\approx$ 31M	DeQA and SAM 2 guidance
MoFE-MLLM [119]	arXiv 2026	32.16	.981	37.86	.982	31.56	.895	32.23	.932	23.40	.866	31.44	.931	$\approx$ 3792M	MLLM-guided frequency routing
QuReC [120]	ACM MM 2026	31.72	.982	39.23	.986	31.56	.891	32.50	.936	24.23	.867	31.85	.932	30M	CLIP prototype matching
MGN-AIR [121]	ACM MM 2026	31.74	.982	39.47	.986	31.12	.884	31.18	.917	23.67	.862	31.44	.926	34M	Pixel-wise restoration matrix
UniRestorer [178]	ICLR 2026	36.44	.984	41.68	.996	31.71	.946	31.48	.948	26.67	.863	33.60	.947	627M	Multi-granularity expert routing

recent studies highlight the importance of supervision quality and knowledge retention. Transferring target-domain weather patterns to clean source images improves adaptation without paired target-domain references [209]. Generated restoration targets also provide useful supervision, but the quality-control experiments show that unfiltered targets yield less consistent improvements [53]. Meanwhile, freezing previously learned restoration paths preserves earlier capabilities during continual adaptation, although incorrect routing still reduces test performance [214]. These findings suggest that practical gains depend on suitable adaptation data and retention of existing restoration knowledge, not training-set expansion alone.

Real-weather data provide an additional acquisition distinction. WeatherStream selects time-multiplexed clear and adverse-weather observations from online camera streams through light-transport-based filtering [281]. WeatherBench

Table XII. Universal restoration results on CDD-11 composite degradations. We report PSNR (↑) and SSIM (↑). '–' denotes an unavailable value. Single-, Double-, and Triple-Deg. report means for one-, two-, and three-order composite degradation subsets. OneRestore uses automatically generated scene descriptors. For most works, # Params. includes the inference components. For methods with unclear configurations, we report the approximate parameter count of the restoration backbone.

Method	Publication	Single-Deg. Avg.		Double-Deg. Avg.		Triple-Deg. Avg.		Overall Avg.		# Params	Key Features
		PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM		
AirNet [203]	CVPR 2022	25.60	.885	22.97	.796	22.02	.717	23.75	.814	9M	Contrastive degradation cues
DGUNet [274]	CVPR 2022	–	–	–	–	–	–	26.92	.856	17M	Generalized unfolding
MIRNet-v2 [273]	TPAMI 2022	28.46	.912	24.11	.818	22.35	.719	25.37	.834	6M	Multi-scale residual blocks
TransWeather [201]	CVPR 2022	24.94	.860	22.42	.764	21.28	.667	23.13	.781	22M	Weather queries
Fourmer [283]	ICML 2023	26.08	.875	22.28	.771	21.06	.681	23.44	.792	1M	Fourier spatial-channel modeling
PromptIR [204]	NeurIPS 2023	28.88	.920	24.46	.834	23.54	.750	25.90	.850	38M	Input-conditioned prompts
SRUDC [284]	ICCV 2023	–	–	–	–	–	–	27.64	.860	7M	Scattering-guided dual branches
WeatherDiff [285]	TPAMI 2023	23.81	.860	21.98	.786	21.14	.707	22.49	.799	83M	Patch diffusion
WGWS-Net [286]	CVPR 2023	29.97	.923	25.76	.850	23.94	.772	26.96	.862	26M	Shared and specific features
CoR-HAIR [221]	arXiv 2024	–	–	–	–	–	–	28.33	.869	30M	Discriminator-guided chaining
CoR-OneRestore [221]	arXiv 2024	–	–	–	–	–	–	28.84	.879	8M	Discriminator-guided chaining
DA-CLIP-NAFNet [8]	ICLR 2024	32.29	.926	27.35	.859	23.14	.732	28.38	.860	–	CLIP content-degradation cues
HAIR [256]	arXiv 2024	–	–	–	–	–	–	27.85	.866	29M	Classifier-guided hypernetwork
InstructIR [3]	ECCV 2024	31.82	.919	26.77	.848	24.58	.769	28.21	.859	–	BGE instruction guidance
OKNet [287]	AAAI 2024	29.31	.926	25.16	.844	23.31	.762	26.33	.859	5M	Omni-kernel network
OneRestore [205]	ECCV 2024	31.68	.938	27.35	.866	24.84	.790	28.47	.878	$\approx$ 18M	Scene descriptors
DCPT-NAFNet [24]	ICLR 2025	34.50	.947	29.33	.881	26.32	.806	30.66	.891	68M	Degradation classification (DCPT)
MIRAGE-S [269]	arXiv 2025	32.79	.944	28.00	.875	25.73	.800	29.33	.886	10M	Manifold contrastive learning
MoCE-IR-S [263]	CVPR 2025	32.54	.941	27.73	.869	25.40	.790	29.06	.881	11M	Complexity experts
VL-UR [116]	arXiv 2025	31.97	.938	27.59	.868	25.25	.791	28.76	.879	–	CLIP weather embeddings
AllRestorer [280]	TPAMI 2026	–	–	–	–	–	–	33.72	.944	12M	CLIP scene descriptors
UniRestorer [178]	ICLR 2026	34.72	.965	29.63	.916	26.43	.890	30.90	.929	364M	Multi-granularity expert routing
MoFE-MLLM [119]	arXiv 2026	34.11	.946	29.11	.878	26.21	.801	30.40	.889	$\approx$ 3792M	MLLM frequency routing
QuReC [120]	ACM MM 2026	34.39	.948	29.44	.883	26.54	.810	30.71	.893	30M	CLIP prototype matching
MGN-AIR [121]	ACM MM 2026	34.57	.946	29.09	.879	26.21	.802	30.56	.889	$\approx$ 34M	CLIP-conditioned pixel matrix

Table XIII. Agentic image restoration results on MiO100. Groups A-C denote the MiO100 degradation subsets used in SkillIR [35]. We report PSNR (↑), LPIPS (↓), and MUSIQ (↑). '–' denotes an unavailable value. OPERA, SEAR, and both HarnessIR variants report Y-channel PSNR. OPERA variants use planning-only optimization or joint planning-execution optimization. EvoIR-Agent uses the AgenticIR (I) or 4KAgent (II) tool pool with fidelity or perceptual preferences. For MiO100, HarnessIR reports ΔMUSIQ rather than absolute MUSIQ, so its MUSIQ entries are omitted.

Method	Publication	MiO100 Group A			MiO100 Group B			MiO100 Group C			Key Features
		PSNR	LPIPS	MUSIQ	PSNR	LPIPS	MUSIQ	PSNR	LPIPS	MUSIQ	
MiOIR [206]	arXiv 2024	20.84	0.3715	47.82	20.56	0.3243	51.87	15.63	0.5376	37.95	Sequential and prompt learning
4KAgent [225]	NeurIPS 2025	21.48	0.3019	63.19	20.95	0.3017	62.69	19.77	0.4271	55.56	Quality-driven MoE, reflection, rollback
AgenticIR [30]	ICLR 2025	21.04	0.3148	56.88	20.55	0.3072	57.57	18.82	0.4493	48.68	Experience-guided scheduling and rollback
MAIR [32]	IJCV 2026	21.02	0.2963	59.19	20.92	0.2788	60.98	19.42	0.4142	51.36	Three-stage, two-level multi-agent restoration
OPERA (planning) [215]	arXiv 2026	22.32	0.3300	59.34	22.28	0.3100	59.10	20.75	0.4700	49.91	GRPO-trained planner, fixed tools
OPERA (joint) [215]	arXiv 2026	24.81	0.2300	57.84	24.63	0.2000	58.41	23.04	0.3000	51.27	Joint planning–execution optimization
TIR-Agent [34]	ECCV 2026	22.07	0.2874	63.69	22.80	0.2725	65.24	19.53	0.4120	56.93	Direct tool-calling policy, SFT then RL
EvoIR-Agent (I, fidelity) [228]	arXiv 2026	23.49	0.2840	–	23.41	0.2753	–	21.52	0.4108	–	Experience retrieval, reflection, rollback
EvoIR-Agent (I, perceptual)	arXiv 2026	–	–	60.10	–	–	60.98	–	–	51.36	Experience retrieval, reflection, rollback
EvoIR-Agent (II, fidelity)	arXiv 2026	23.86	0.3085	54.37	23.38	0.3117	53.28	22.12	0.4528	41.36	Experience retrieval, reflection, rollback
EvoIR-Agent (II, perceptual)	arXiv 2026	22.31	0.2831	61.82	22.52	0.2694	61.96	20.69	0.4072	55.21	Experience retrieval, reflection, rollback
SEAR [216]	ECCV 2026	21.80	0.3019	61.29	22.13	0.2890	60.65	20.61	0.4173	52.73	Trajectory retrieval and pruning-aware MCTS
SkillIR [35]	arXiv 2026	22.39	0.2744	62.31	21.73	0.2655	62.60	20.59	0.4031	54.57	Scene-aware skills, transition verification
HarnessIR-GPT [36]	arXiv 2026	20.19	0.2800	–	19.96	0.2800	–	18.95	0.3400	–	Prompt composition and verified refinement
HarnessIR-NB2 [36]	arXiv 2026	22.45	0.2000	–	22.94	0.1900	–	21.23	0.2500	–	Prompt composition and verified refinement

instead photographs controlled physical rain, snow, and haze through sequential clean and degraded capture [282]. The former scales selection from naturally observed weather, whereas the latter increases control over paired acquisition. Neither should be described as simultaneous capture of an identical clean and degraded scene. Coverage of several weather categories also does not establish evaluation on every simultaneous degradation combination.

Generalization Performance Comparison. Unseen-task experiments show that examples can provide restoration guidance without parameter updates. The results of Hidden-Shot on tasks excluded from training support using demonstration pairs and task descriptions to specify new restoration tasks [183]. However, adaptation is still used in some settings. For example, MPerceiver reports direct transfer to some image domains and few-shot fine-tuning for other restoration tasks [147]. Pretrained models therefore support distinct forms of transfer through conditioning or parameter adaptation. Their practical generalization depends on whether the available examples or training data supply the information needed for the target task.

C. Agentic Image Restoration

Table XIII indicates that restoration quality depends on both planning and the tools used to execute a plan. The planner-only

and joint-training comparisons of OPERA support adapting restoration tools to their role in the selected sequence [215]. SkillIR further associates scene-aware skills and candidate verification with restoration quality, while dynamic skills also affect the number of tool calls [35]. These findings highlight the importance of selecting suitable operations and preventing unsuccessful intermediate results from propagating. However, fewer restoration calls do not necessarily mean shorter runtime, since perception, retrieval, and verification also require computation. The relevant comparison is therefore the quality and cost of the complete system, not the controller or tool count alone.

On generalization, the results of Restore-R1 on held-out degradation combinations support transferring learned tool-selection policies to new combinations of degradations [38]. This is evidence for adapting operation sequences within an existing tool set, not for solving entirely new restoration tasks. User feedback provides a separate way to revise tool choices and lighting instructions, as illustrated by Clarity ChatGPT [226] and VLM-IMI [227]. Their interactive examples demonstrate controllability, but do not establish generalization to unseen degradation combinations. The available evidence therefore supports flexibility in the restoration process more clearly than unrestricted task generalization.

VI. Future Directions

A. Generalization and Adaptation to Real Degradations

Generalization to real-world degradations depends on the coverage and quality of pretraining data. Large datasets may still under-represent rare degradations and complex combinations, while the quality of target images affects what the model learns to reconstruct. FoundIR-v2 addresses these issues through adaptive sampling and multimodal target filtering [184]. Future work should investigate how data sampling, target selection, and synthetic degradation design improve performance on under-represented cases, rather than only increasing average performance. Ablation studies with the same architecture and training budget can help identify the contribution of each component.

Even with broader training coverage, real-world inputs can violate the degradation assumptions used by a restoration method. BlindDPS estimates unknown degradations but still relies on such assumptions [142]. When they do not match the actual degradation, more accurate parameter estimates alone may be insufficient. Future work should investigate how to detect these mismatches and determine whether to refine the degradation model, select a suitable restorer, or adapt an existing model. GRIDS selects among fixed restorers [208], whereas AdaptIR introduces trainable parameters for new tasks [26]. These approaches motivate studying when existing capabilities can be reused, when additional modules are needed, and when shared model weights require updating. The choice should account for adaptation data and storage requirements, as well as the risk of degrading performance on previously learned tasks.

Evaluating these capabilities requires separating generalization from improvements obtained through additional adaptation. New degradation combinations, changes in image domains, and degradation types excluded from training should be evaluated separately. Within each setting, comparisons should distinguish models used without adaptation from those receiving additional examples or parameter updates at deployment. Adaptation methods should be compared under matched data constraints, including access to paired targets and examples from earlier tasks. This would clarify which gains arise from pretrained capabilities and which depend on additional data and model updates.

B. Faithful and User-Controllable Restoration

Improving perceptual quality should preserve important content, including text, identity, and geometry, while avoiding details unsupported by the input. Under certain inverse-problem assumptions, multiple reconstructions can fit the same observations [288], [289], making accurate and stable recovery difficult. Uncertainty estimates may identify details weakly constrained by the observations, but high uncertainty does not necessarily indicate an error [290]. Existing uncertainty-quantification methods also rely on assumptions that may fail under real-world degradations [291]. Future work should investigate how these estimates can constrain the influence of generative priors and whether this reduces unsupported details without increasing residual degradation. A useful direction is to make this control sensitive to local evidence, so that uncertain regions do not receive the same treatment as well-observed structures.

User instructions make this preservation requirement more specific by identifying what should change and what should remain unchanged. Pixel-level, semantic, and region-specific controls provide interfaces for expressing these requests [155], [164]. However, following a preference does not establish that generated details match the original scene. Future systems should distinguish requests to recover existing content from requests to modify its appearance, resolve conflicting instructions, and explain when the input provides insufficient evidence for reliable reconstruction. Evaluation should assess both adherence to the requested changes and preservation of content outside their scope.

C. Reliable Assessment for Agentic Restoration

Restoration agents need reliable feedback to determine whether processing removes degradation or introduces new errors. Joint quality-hallucination assessment [174] and artifact localization [175] offer complementary evidence. Future work should connect quality judgments to specific defects and regions, validating scores, explanations, and localization separately across methods and domains. The feedback should also indicate when the evidence cannot support a confident decision. This evidence should guide decisions to retain results, select another tool, or stop. SkillIR verifies candidates [35], while DiTTo adapts to new experts [224]. Learning from successful and failed attempts may improve action selection. Comparisons should hold tools and computation budgets constant, with final outputs evaluated independently of scores used for training or selection.

D. Efficient and Reproducible Restoration Agents

Reducing redundant assessment, planning, and candidate search is important for efficient restoration agents. Future work could investigate when intermediate assessments or plans can be reused without compromising restoration quality. Efficiency comparisons should include execution and weigh online savings against offline data generation, training, and adaptation costs. Reported savings should not omit repeated tool calls. Preparation and deployment costs should be reported separately, specifying inference components and pretraining and adaptation data [183]. Latency measurements require consistent hardware and image settings. For agents with stored experience, reproducibility also depends on prompts, tool versions, initial memory, and image order. Recording these settings and repeating runs would reveal variation in quality, content errors, and cost.

VII. Conclusion

Foundation models offer pretrained guidance, adaptable restorers, and agent control. Reported restoration gains depend on the backbone and deployment conditions. Training-task coverage, fixed-parameter transfer, and adaptation demonstrate different capabilities. Agentic evaluation must also consider feedback reliability and full system cost, not tool counts alone. Future progress should preserve content while supporting user control and improve the reliability, efficiency, and reproducibility of restoration agents.

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